

Supplement to “Multiscale Comparison of Nonparametric Trend Curves”

A Simulations

In this section, we investigate our test and clustering methods by means of Monte Carlo experiments.

A.1 Size and power of the multiscale test

The simulation design is as follows: We generate data from the model $Y_{it} = m_i(\frac{t}{T}) + \beta_i^\top \mathbf{X}_{it} + \alpha_i + \varepsilon_{it}$, where the number of time series is set to $n = 15$ and we consider different time series lengths T . We include 3 covariates and model them by the VAR(3) process

$$\mathbf{X}_{it} = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix} \mathbf{X}_{i,t-1} + \boldsymbol{\nu}_{it},$$

where $\mathbf{X}_{it} = (X_{it,1}, X_{it,2}, X_{it,3})^\top$, $\boldsymbol{\nu}_{it} = (\nu_{it,1}, \nu_{it,2}, \nu_{it,3})^\top$ and we choose $a_1 = a_2 = a_3 = 0.25$.

The innovations $\boldsymbol{\nu}_{it}$ are drawn i.i.d. from a multivariate normal distribution $N(0, \Phi)$ with

$$\Phi = \begin{pmatrix} 1 & \varphi & \varphi \\ \varphi & 1 & \varphi \\ \varphi & \varphi & 1 \end{pmatrix},$$

where $\varphi = 0.25$ specifies the correlation between the 3 covariates. The parameter vector $\beta_i = (\beta_{i,1}, \beta_{i,2}, \beta_{i,3})^\top$ is chosen to equal $(1, 1, 1)^\top$ for each i . Moreover, for each i , the idiosyncratic error process $\{\varepsilon_{it} : 1 \leq t \leq T\}$ follows the AR(1) model $\varepsilon_{it} = a\varepsilon_{it-1} + \eta_{it}$, where $a = 0.25$ and the innovations η_{it} are i.i.d. normally distributed with mean 0 and standard deviation 0.25. The vector of fixed effects $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)^\top$ is modelled as a

normally distributed random vector. In particular, $\boldsymbol{\alpha} \sim N(0, \Sigma)$ with

$$\Sigma = \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho \\ \rho & \cdots & \rho & 1 \end{pmatrix},$$

where $\rho = 0.25$ gives the correlation across i . To generate data under the null $H_0 : m_1 = \dots = m_n$, we let $m_i = 0$ for all i without loss of generality. To produce data under the alternative, we use the bump function

$$m_1(u) = b \cdot \vartheta(u, 0.3, 0.1) - b \cdot \vartheta(u, 0.7, 0.1) \quad (\text{A.1})$$

with $\vartheta(u, c, d) = \mathbb{1} \left\{ \frac{|u-c|}{d} \leq 1 \right\} \left(1 - \frac{(u-c)^2}{d^2} \right)^2$ for $b \in \{0.25, 0.5, 0.75\}$ (depicted in Figure A.1) and let $m_i = 0$ for $i \neq 1$. Note that the normalisation constraint $\int_0^1 m_1(u) du = 0$ is directly satisfied in this case. For each simulation exercise, we draw 5000 samples.

Figure A.1 illustrates that the described simulation setup is quite challenging already without covariates and fixed effects. The black lines in the plot show the bump function m_1 for different heights of the bump $b \in \{0, 0.25, 0.5, 0.75\}$, where the value $b = 0$ corresponds to the null H_0 . The red lines show one realization of the time series $\{Y_{1t}^\circ : 1 \leq t \leq T\}$ for $T = 250$, where $Y_{1t}^\circ = m_1(\frac{t}{T}) + \varepsilon_{1t}$ is an idealized version of Y_{1t} without covariates and fixed effects. The values $b = 0.25, 0.5, 0.75$ correspond to three different alternatives. In particular, the larger the bump size b , the further away from the null is the corresponding alternative. For the smallest value $b = 0.25$, the bump function is very difficult to detect since the noise level is very high, or put differently, since the signal-to-noise ratio is very small (as is clearly visible in the upper right panel of Figure A.1). For $b = 0.5$ and $b = 0.75$, the signal-to-noise ratio slowly gets bigger, making it less difficult to detect the bump function. Nevertheless, even for $b = 0.75$, there is substantial noise around the bump function.

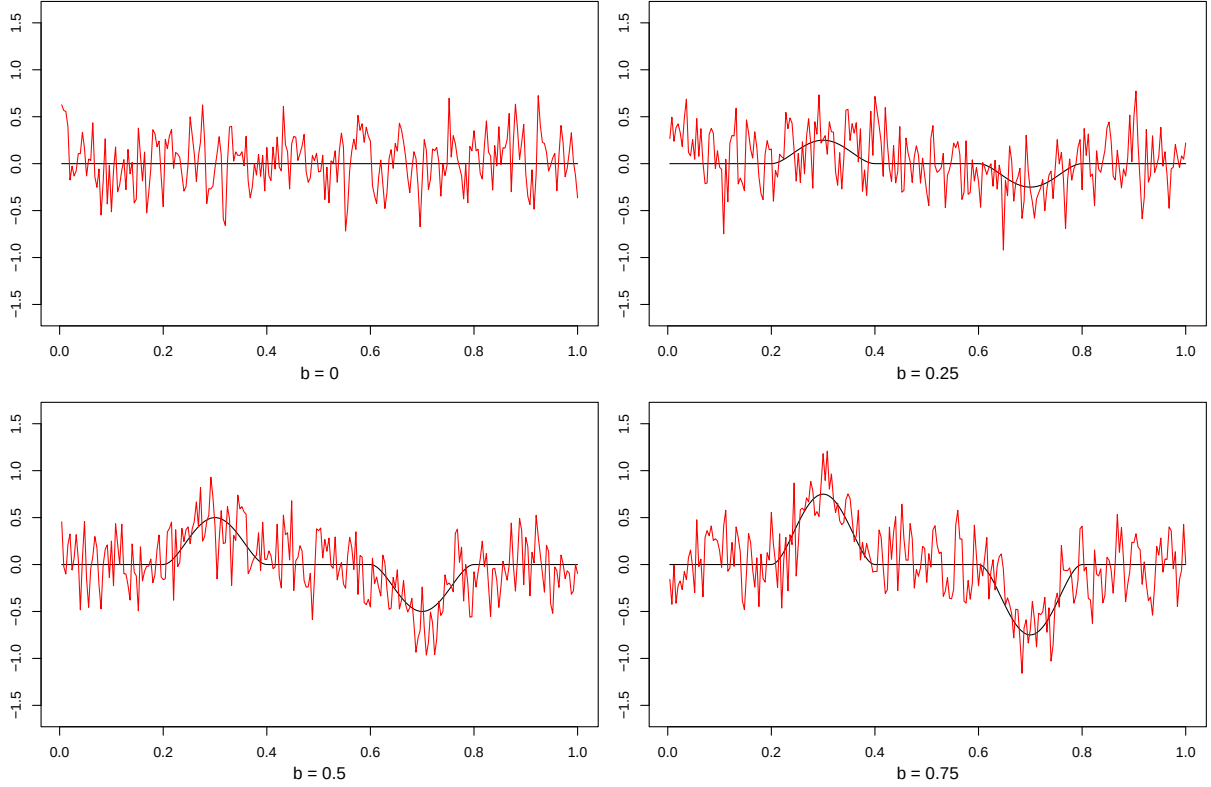


Figure A.1: In black: the bump function m_1 for different bump heights $b \in \{0, 0.25, 0.5, 0.75\}$ with $b = 0$ corresponding to the null H_0 . In red: the time series $\{Y_{1t}^o : 1 \leq t \leq T\}$ with $Y_{1t}^o = m_1(t/T) + \varepsilon_{1t}$ for one simulation run and $T = 250$.

Our multiscale test is implemented as follows: The estimators $\hat{\beta}_i$ and $\hat{\alpha}_i$ of the unknown parameters β_i and α_i are computed as described in Section 3.1. The long-run error variance is estimated by the difference-based estimator proposed in Khismatullina and Vogt (2020), where the tuning parameters q and r are set to 25 and 10, respectively. In order to construct our test statistics, we use an Epanechnikov kernel and the location-scale grid $\mathcal{G}_T = U_T \times H_T$ with

$$U_T = \{u \in [0, 1] : u = \frac{5t}{T} \text{ with } 1 \leq t \leq T\} \quad (\text{A.2})$$

$$H_T = \{h \in [\frac{\log T}{T}, \frac{1}{4}] : h = \frac{5t-3}{T} \text{ with } 1 \leq t \leq T\}. \quad (\text{A.3})$$

We thus consider intervals $\mathcal{I}_{u,h} = [u - h, u + h]$ which contain 5, 15, 25, ... data points. The critical value $q_{n,T}(\alpha)$ of our test is computed by Monte Carlo methods as described in Section 3.4, where we set $L = 5000$.

Table A.1: Empirical size and power of the multiscale test for different time series lengths T and nominal sizes α . Each panel gives the results for a different bump height b , where $b = 0$ corresponds to the null and $b = 0.25, 0.5, 0.75$ to three different alternatives.

(a) $b = 0$				(b) $b = 0.25$			
nominal size α				nominal size α			
T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.006	0.035	0.070	100	0.011	0.046	0.095
250	0.020	0.079	0.134	250	0.158	0.348	0.473
500	0.012	0.058	0.112	500	0.524	0.762	0.854

(c) $b = 0.5$				(d) $b = 0.75$			
nominal size α				nominal size α			
T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.012	0.058	0.105	100	0.010	0.035	0.071
250	0.837	0.952	0.979	250	0.967	0.992	0.996
500	1.000	1.000	1.000	500	1.000	1.000	1.000

Table A.1 reports the simulation results, in particular, the empirical size and power of the multiscale test. Both the empirical size and the empirical power are computed as the number of simulations in which the test rejects the global null H_0 divided by the total number of simulations. The four panels (a)–(d) of Table A.1 correspond to different values of the bump size b . Panel (a) corresponds to the case $b = 0$ and thus gives the empirical size of the test under the null H_0 . Panels (b)–(d), in contrast, report the empirical power against the three alternatives with $b \in \{0.25, 0.5, 0.75\}$. Inspecting panel (a), one can see that the empirical size of the test gives a reasonable approximation to the target α in all scenarios under investigation, even though the size numbers are a bit downward/upward biased for the smaller time series lengths $T = 100$ and $T = 250$. Inspecting panels (b)–(d), one can further see that the test has good power properties. For the smallest time series length $T = 100$, the power numbers are still very low, reflecting the fact that the noise level in the data is extremely high, even for the largest bump size $b = 0.75$. However, as we increase the time series length T , the power increases quickly. For $T = 250$ and $b = 0.5$, for instance, we already have power of over 95% when $\alpha \geq 0.05$.

A.2 Robustness checks on the choice of tuning parameters

The main tuning parameter of our method is the grid $\mathcal{G}_T$, which in particular specifies the bandwidths that are taken into account. In order to assess the robustness of our test with respect to the grid, we repeat the simulation exercises from Section A.1 with two different choices of the grid: a sparser and a finer grid than the one used in Section A.1. The sparser grid is defined as

$$\mathcal{G}_T^{\text{sparse}} = \left\{ (u, h) \subseteq [0, 1] : (u, h) = ((2s+1)h, h) \text{ for } s = 0, \dots, \left\lfloor \frac{h^{-1} - 1}{2} \right\rfloor \right. \\ \left. \text{and } h \in H_T^{\text{sparse}} \right\} \quad (\text{A.4})$$

with $H_T^{\text{sparse}} = \{h = 2^k h_{\min} \text{ for } k = 0, \dots, K\}$, where $h_{\min} = \frac{\lceil \log T \rceil}{T}$ and K is such that $2^K h_{\min} \leq \frac{1}{4}$, i.e., $K \leq \lfloor \log(\frac{T}{4 \lceil \log T \rceil}) \frac{1}{\log(2)} \rfloor$. This is a dyadic scheme with scales in the set H_T^{sparse} as commonly used in Wavelet analysis. The finer grid is defined as $\mathcal{G}_T^{\text{fine}} = U_T^{\text{fine}} \times H_T^{\text{fine}}$, where $U_T^{\text{fine}} = \{u \in [0, 1] : u = \frac{t}{T} \text{ with } 1 \leq t \leq T\}$ and $H_T^{\text{fine}} = \{h \in [\frac{\log T}{T}, \frac{1}{4}] : h = \frac{2t}{T} \text{ with } 1 \leq t \leq T\}$. Apart from the choice of grid, we work with precisely the same simulation setup as in Section A.1.¹

The simulation results for the three grids are reported in Table A.2.² Each column corresponds to a particular choice of grid. Panel (a) in each column specifies the empirical size of the test, while panels (b)–(d) give the empirical power against the alternatives with bump size $b \in \{0.25, 0.5, 0.75\}$. As can be seen, the size numbers in Table A.2 are fairly stable across the three different grids. In particular, we get a reasonable approximation to the target size α , no matter which of the three grids we are working with. Moreover, the power numbers for $\mathcal{G}_T$ and the finer grid $\mathcal{G}_T^{\text{fine}}$ are quite similar to each other. Those

¹As the grid $\mathcal{G}_T^{\text{fine}}$ is very fine, the number of tests to be carried out is extremely large, which significantly increases the computational burden. To decrease computational time, we proceed as follows when using the fine grid $\mathcal{G}_T^{\text{fine}}$: we calculate the Gaussian quantile based on $L = 1000$ (rather than 5000) bootstrap samples and we simulate 1000 samples (rather than 5000) for each simulation scenario.

²Note that the simulation results for the grid $\mathcal{G}_T$ in the second column of Table A.2 are identical with those in Table A.1. For ease of comparison, we have repeated these results as part of Table A.2.

Table A.2: Empirical size and power of the multiscale test for three different grids. Each column corresponds to a particular choice of grid. Panels (a)–(d) of each column have the same structure as in Table A.1.

sparser grid $\mathcal{G}_T^{\text{sparse}}$				grid $\mathcal{G}_T$ from Section A.1				finer grid $\mathcal{G}_T^{\text{fine}}$			
(a) $b = 0$				(a) $b = 0$				(a) $b = 0$			
nominal size α				nominal size α				nominal size α			
T	0.01	0.05	0.1	T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.008	0.036	0.070	100	0.006	0.035	0.070	100	0.010	0.031	0.054
250	0.014	0.055	0.100	250	0.020	0.079	0.134	250	0.014	0.050	0.106
500	0.013	0.055	0.109	500	0.012	0.058	0.112	500	0.012	0.054	0.117
(b) $b = 0.25$				(b) $b = 0.25$				(b) $b = 0.25$			
nominal size α				nominal size α				nominal size α			
T	0.01	0.05	0.1	T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.008	0.042	0.082	100	0.011	0.046	0.095	100	0.014	0.044	0.075
250	0.078	0.174	0.261	250	0.158	0.348	0.473	250	0.129	0.283	0.389
500	0.203	0.438	0.576	500	0.524	0.762	0.854	500	0.526	0.761	0.824
(c) $b = 0.5$				(c) $b = 0.5$				(c) $b = 0.5$			
nominal size α				nominal size α				nominal size α			
T	0.01	0.05	0.1	T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.011	0.047	0.100	100	0.012	0.058	0.105	100	0.019	0.053	0.091
250	0.501	0.709	0.807	250	0.837	0.952	0.979	250	0.822	0.935	0.965
500	0.995	0.999	0.999	500	1.000	1.000	1.000	500	1.000	1.000	1.000
(d) $b = 0.75$				(d) $b = 0.75$				(d) $b = 0.75$			
nominal size α				nominal size α				nominal size α			
T	0.01	0.05	0.1	T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.007	0.027	0.064	100	0.010	0.035	0.071	100	0.009	0.031	0.052
250	0.743	0.893	0.946	250	0.967	0.992	0.996	250	0.967	0.992	0.997
500	1.000	1.000	1.000	500	1.000	1.000	1.000	500	1.000	1.000	1.000

for the sparser grid $\mathcal{G}_T^{\text{sparse}}$, in contrast, are somewhat lower, which may be explained as follows: As $\mathcal{G}_T^{\text{sparse}}$ is considerably sparser than the other two grids, much less intervals $\mathcal{I}_{u,h} = [u - h, u + h]$ are taken into account. In particular, intervals may potentially be ignored where the deviation from the null is quite strong. This results in a loss of power. Taken together, our simulation results suggest the following: (i) The particular choice of grid has a negligible effect on the empirical size of the test. (ii) As long as the grid is sufficiently rich, the choice of grid also has a negligible effect on the power of the test.

Apart from the grid, our method depends on the following tuning parameters: (i) the number of bootstrap samples L to compute the Gaussian quantile, (ii) the kernel K , and (iii) secondary tuning parameters for the computation of the long-run error variance. As long as L is chosen large enough (say $L \geq 1000$), the exact value of L should have a negligible effect. We use $L = 5000$ throughout the paper.³ As a robustness check, we have rerun everything with $L = 1000$, which (as expected) yields almost identical results.⁴ As suggested by classical nonparametric theory, the Epanechnikov kernel (which is used throughout the paper) has good properties and the choice of kernel K is of minor importance, in particular, much less important than the choice of bandwidth. Thus, we have not carried out robustness checks with respect to the choice of kernel. Our procedure can be combined with any method off-the-shelf for long-run error variance estimation. In the current simulation study, we work with the long-run variance estimator from Khismatullina and Vogt (2020), where extensive robustness checks with respect to the choice of tuning parameters have been carried out. We refer to Section 5.2 therein for the details.

A.3 Performance of the test for large n

So far, we have fixed the number of time series at $n = 15$. We now examine the performance of our test as n gets larger. To do so, we rerun the simulation exercises from Section A.1 for $n \in \{25, 50, 100\}$ and T as before, i.e., $T \in \{100, 250, 500\}$. We work with precisely the same simulation setup as specified in Section A.1 except for the fact that we use the smaller grid $\mathcal{G}_T^{\text{sparse}}$ defined in (A.4) to decrease the computational burden. Overall, our simulation results suggest that the test performs well for larger n . The results are presented in Table A.3, with each column corresponding to a particular choice of n . The upper panels (a) of Table A.3 show that the empirical size of the test provides a reasonably good approximation to the target α across all scenarios under consideration. In particular, the size numbers

³Except for the simulations with the fine grid $\mathcal{G}_T^{\text{fine}}$, where $L = 1000$.

⁴The results are not reported here but are available from the authors upon request.

Table A.3: Empirical size and power of the multiscale test. Each column corresponds to a particular choice of n . Panels (a)–(d) of each column have the same structure as in Table A.1.

$n = 15$			$n = 25$			$n = 50$			$n = 100$		
(a) $b = 0$			(a) $b = 0$			(a) $b = 0$			(a) $b = 0$		
nominal size α			nominal size α			nominal size α			nominal size α		
T	0.01	0.05	0.1	T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.008	0.036	0.070	100	0.005	0.032	0.065	100	0.005	0.034	0.066
250	0.014	0.055	0.100	250	0.012	0.054	0.098	250	0.015	0.061	0.110
500	0.013	0.055	0.109	500	0.018	0.059	0.108	500	0.010	0.068	0.128
(b) $b = 0.25$			(b) $b = 0.25$			(b) $b = 0.25$			(b) $b = 0.25$		
nominal size α			nominal size α			nominal size α			nominal size α		
T	0.01	0.05	0.1	T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.008	0.042	0.082	100	0.008	0.042	0.080	100	0.008	0.039	0.078
250	0.078	0.174	0.261	250	0.056	0.159	0.237	250	0.046	0.134	0.215
500	0.203	0.438	0.576	500	0.181	0.378	0.512	500	0.106	0.329	0.453
(c) $b = 0.5$			(c) $b = 0.5$			(c) $b = 0.5$			(c) $b = 0.5$		
nominal size α			nominal size α			nominal size α			nominal size α		
T	0.01	0.05	0.1	T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.011	0.047	0.100	100	0.008	0.045	0.089	100	0.008	0.041	0.077
250	0.501	0.709	0.807	250	0.410	0.634	0.739	250	0.364	0.589	0.691
500	0.995	0.999	0.999	500	0.994	0.999	1.000	500	0.983	0.999	1.000
(d) $b = 0.75$			(d) $b = 0.75$			(d) $b = 0.75$			(d) $b = 0.75$		
nominal size α			nominal size α			nominal size α			nominal size α		
T	0.01	0.05	0.1	T	0.01	0.05	0.1	T	0.01	0.05	0.1
100	0.007	0.027	0.064	100	0.005	0.032	0.062	100	0.006	0.033	0.063
250	0.743	0.893	0.946	250	0.656	0.848	0.907	250	0.615	0.798	0.874
500	1.000	1.000	1.000	500	1.000	1.000	1.000	500	1.000	1.000	1.000

remain fairly stable as n increases, even though they appear to get a little less precise for larger n . Panels (b)–(d) further show that the test has reasonable power across all considered scenarios. The power numbers, however, decrease somewhat as n gets larger, which may be explained as follows: Let $\mathcal{M} = \{(u, h, i, j) : (u, h) \in \mathcal{G}_T \text{ and } 1 \leq i < j \leq n\}$ be the collection of all location-scale points (u, h) and all pairs of time series (i, j) under consideration and let $\mathcal{M}_1 \subseteq \mathcal{M}$ be the set of tuples (u, h, i, j) for which the local null hypothesis $H_0^{[i,j]}(u, h)$ is false. Since in the simulation setup at hand, all trend functions are the same except the first one (m_1 being the bump function from (A.1) with bump size b and $m_i = 0$ for all $i \neq 1$), it holds that $|\mathcal{M}_1|/|\mathcal{M}| \rightarrow 0$ quite fast as $n \rightarrow \infty$ (where $|\mathcal{M}|$ denotes the cardinality of the set $\mathcal{M}$). Hence, as n increases, the total number of hypotheses to be tested grows much faster than the number of hypotheses which are violated. As a consequence, it gets harder to detect these violations, which is reflected in somewhat lower power numbers.

A.4 Comparison of our test with SiZer

To the best of our knowledge, the only other multiscale test for comparing trend curves available in the literature is the SiZer test developed in Park et al. (2009). In this section, we compare our method with this test. Notably, Park et al. (2009) consider a simplified version of our model without covariates and fixed effects. Moreover, rigorous theory has been developed only for the case of $n = 2$ time series. The authors propose a possible extension of their approach to the case of more than $n = 2$ time series, but the extension does not allow for a pairwise comparison of the time series. Moreover, it is not clear how to calculate the actual size and power of the test in the case of more than $n = 2$ time series. For our comparison, we thus consider the model $Y_{it} = m_i\left(\frac{t}{T}\right) + \varepsilon_{it}$ without covariates and fixed effects and we choose the number of time series to be equal to $n = 2$.

Table A.4: Size and power comparison of our multiscale test $\mathcal{T}_{\text{MS}}$ and the SiZer test $\mathcal{T}_{\text{SiZer}}$ from Park et al. (2009).

(a) $b = 0$						
	$\alpha = 0.01$		$\alpha = 0.05$		$\alpha = 0.1$	
	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$
$T = 100$	0.010	0.099	0.047	0.332	0.106	0.524
$T = 250$	0.008	0.162	0.041	0.510	0.102	0.699
$T = 500$	0.009	0.214	0.044	0.598	0.089	0.787

(b) $b = 0.25$						
	$\alpha = 0.01$		$\alpha = 0.05$		$\alpha = 0.1$	
	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$
$T = 100$	0.075	0.251	0.193	0.551	0.317	0.714
$T = 250$	0.253	0.629	0.486	0.876	0.633	0.951
$T = 500$	0.640	0.905	0.817	0.985	0.887	0.996

(c) $b = 0.5$						
	$\alpha = 0.01$		$\alpha = 0.05$		$\alpha = 0.1$	
	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{SiZer}}$
$T = 100$	0.482	0.733	0.703	0.922	0.815	0.967
$T = 250$	0.966	0.998	0.994	1.000	0.998	1.000
$T = 500$	1.000	1.000	1.000	1.000	1.000	1.000

The simulation setup is as specified in Section A.1 (without covariates and fixed effects, i.e., $\beta_i = 0$ and $\alpha_i = 0$ for all i). As before, to generate data under the null $H_0 : m_1 = m_2$, we let $m_i = 0$ for $i = 1, 2$, and to produce data under the alternative, we use the bump function m_1 defined in (A.1) with bump size $b \in \{0.25, 0.5\}$ while setting $m_2 = 0$. We implement our multiscale test as in Section A.1. For the implementation of the SiZer test, we use the same location-scale grid $\mathcal{G}_T$ as for our test. While our method depends on the long-run error variance σ^2 , the SiZer test depends on the whole autocovariance function $\{\gamma(k) : k \in \mathbb{Z}\}$ of the error process. To make the comparison as fair as possible, we assume the autocovariance function $\gamma(\cdot)$ and thus the long-run error variance $\sigma^2 = \sum_{k=-\infty}^{\infty} \gamma(k)$ to be known when implementing the two methods.

In what follows, we denote our multiscale test by $\mathcal{T}_{\text{MS}}$ and the SiZer test of Park et al. (2009) by $\mathcal{T}_{\text{SiZer}}$. Table A.4 presents the results of our comparison study. Panel (a) compares

the empirical size of the two tests, while panels (b) and (c) provide a power comparison. There is an important difference between $\mathcal{T}_{\text{MS}}$ and $\mathcal{T}_{\text{SiZer}}$ to be noticed: $\mathcal{T}_{\text{MS}}$ is a global test procedure, whereas $\mathcal{T}_{\text{SiZer}}$ is scale-wise in essence. This means that $\mathcal{T}_{\text{MS}}$ tests the hypotheses $H_0^{[1,2]}(u, h)$ simultaneously for all locations $u \in U_T$ and scales $h \in H_T$. In particular, it controls the size simultaneously over both locations u and scales h . $\mathcal{T}_{\text{SiZer}}$, in contrast, tests $H_0^{[1,2]}(u, h)$ simultaneously for all $u \in U_T$ but separately for each scale $h \in H_T$ and, hence, controls the size for each scale $h \in H_T$ separately. This results in $\mathcal{T}_{\text{SiZer}}$ being much too liberal, as panel (a) illustrates: whereas the empirical size numbers of $\mathcal{T}_{\text{MS}}$ closely align with the target size α , the size numbers of $\mathcal{T}_{\text{SiZer}}$ are much larger than the target α . In particular, they deviate more and more strongly from the target α as T increases, since the number of scales h in the grid $\mathcal{G}_T$ increases with T . To summarize, as expected, the global test $\mathcal{T}_{\text{MS}}$ holds the size reasonably well, whereas the scale-wise method $\mathcal{T}_{\text{SiZer}}$ is much too liberal. Inspecting the power numbers in panels (b) and (c), one can further see that in all simulation scenarios, $\mathcal{T}_{\text{SiZer}}$ is more powerful than $\mathcal{T}_{\text{MS}}$. These higher power numbers, however, should not be taken at face value. In particular, they do not imply that $\mathcal{T}_{\text{SiZer}}$ is better in detecting deviations from the null than our test. They are rather a simple consequence of the fact that $\mathcal{T}_{\text{SiZer}}$ is much too liberal as illustrated in panel (a).

There are of course other procedures besides the SiZer test of Park et al. (2009) that can be regarded as potential competitors of our test. The main merit of our method is that we can perform inference uniformly over all locations u and bandwidths (or scales) h in the grid $\mathcal{G}_T$ and over all pairs of time series (i, j) . The potential competitors we are aware of do not have this uniformity property. Specifically, they are not uniform over bandwidths h and usually also not over pairs of time series (i, j) . Hence, most probably, we will see a very similar picture when comparing these methods with ours: they will be much too liberal. In this sense, the above comparison with SiZer is prototypical.

A.5 Performance of the clustering algorithm

We next investigate the finite sample performance of the clustering algorithm from Section 5. To do so, we consider the same simulation setup as in Section A.1 with one exception: we partition the $n = 15$ time series into $N = 3$ groups, each containing 5 time series. Specifically, we set $G_1 = \{1, \dots, 5\}$, $G_2 = \{6, \dots, 10\}$ and $G_3 = \{11, \dots, 15\}$, and we assume that $m_i = f_l$ for all $i \in G_l$ and all $l = 1, 2, 3$. The group-specific trend functions f_1 , f_2 and f_3 are defined as $f_1(u) = 0$, $f_2(u) = 0.35 \cdot \vartheta(u, 0.25, 0.25) - 0.35 \cdot \vartheta(u, 0.75, 0.25)$ and $f_3(u) = \vartheta(u, 0.75, 0.025) - \vartheta(u, 0.25, 0.025)$ with $\vartheta(u, c, d) = \mathbb{1}\left\{\frac{|u-c|}{d} \leq 1\right\} \left(1 - \frac{(u-c)^2}{d^2}\right)^2$ (depicted in Figure A.2). In order to estimate the groups G_1 , G_2 , G_3 and their number $N = 3$, we use the same implementation as in Section A.1 followed by the clustering procedure from Section 5.

The simulation results are reported in Table A.5. The entries in Table A.5(a) are computed as the number of simulations for which $\hat{N} = N$ divided by the total number of simulations. They thus specify the empirical probabilities with which the estimate $\hat{N}$ is equal to the true number of groups $N = 3$. Analogously, the entries of Table A.5(b) give the empirical probabilities with which the estimated group structure $\{\hat{G}_1, \dots, \hat{G}_{\hat{N}}\}$ equals the true one $\{G_1, G_2, G_3\}$. The results in Table A.5 nicely illustrate the theoretical properties of our clustering algorithm. According to Proposition 5.1, the probability that $\hat{N} = N$ and that $\{\hat{G}_1, \dots, \hat{G}_{\hat{N}}\} = \{G_1, G_2, G_3\}$ should be at least $(1 - \alpha)$ asymptotically. For the largest sample size $T = 500$, the empirical probabilities reported in Table A.5 can indeed be seen to (almost) exceed the value $(1 - \alpha)$ for $\alpha \in \{0.05, 0.1\}$ as predicted by Proposition 5.1. For the smaller sample sizes $T = 100$ and $T = 250$, in contrast, the empirical probabilities are substantially smaller than $(1 - \alpha)$. This reflects the asymptotic nature of Proposition 5.1 and is not very surprising. It simply mirrors the fact that for the smaller sample sizes $T = 100$ and $T = 250$, the noise level in the simulated data is very high.

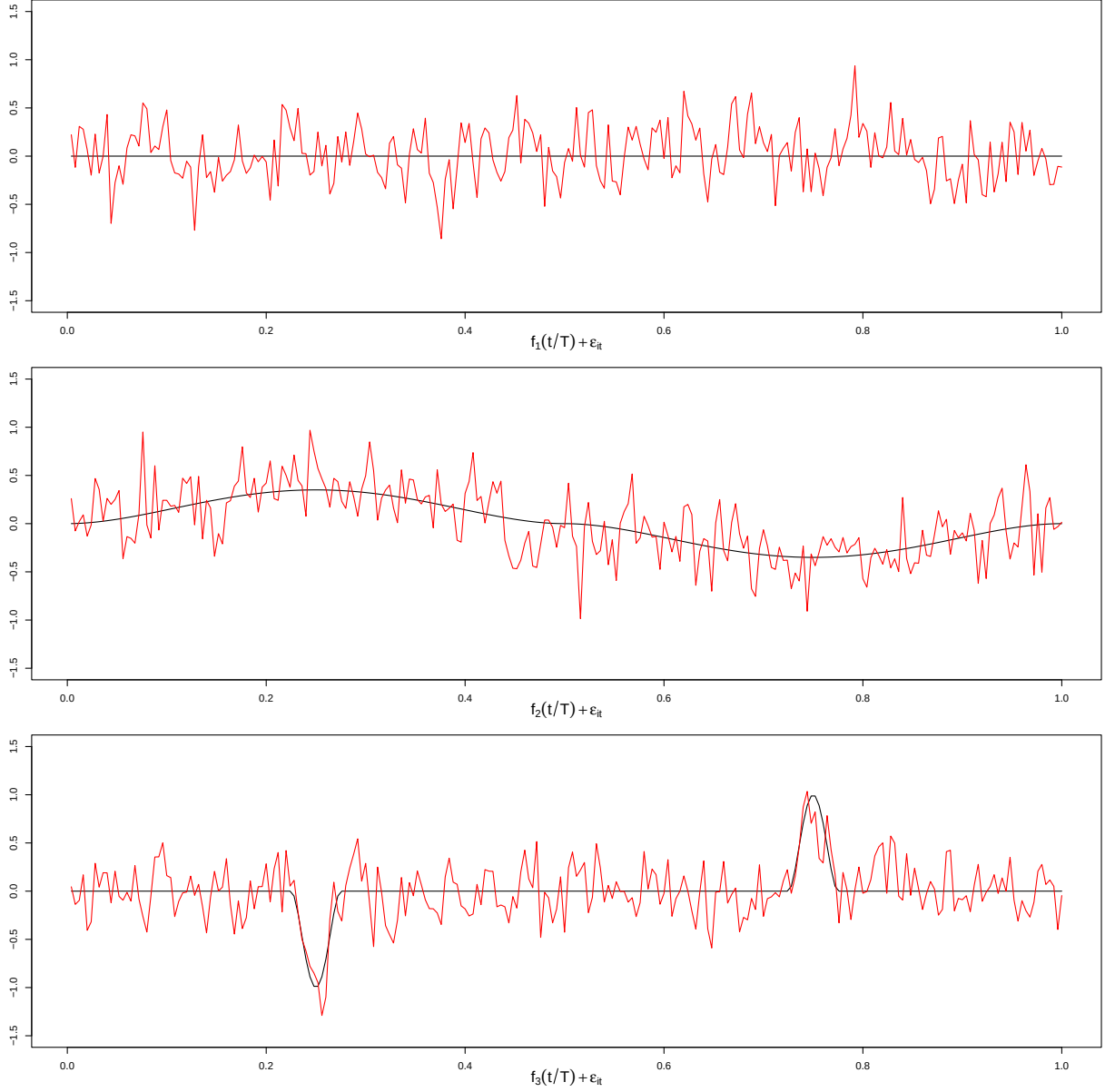


Figure A.2: In black: the group-specific trend functions f_l ($l = 1, 2, 3$). In red: the time series $\{Y_{it}^\circ : 1 \leq t \leq T\}$ with $Y_{it}^\circ = f_l(t/T) + \varepsilon_{it}$ ($l = 1, 2, 3$) for one simulation run and $T = 250$.

Figures A.3 and A.4 give more insight into what happens for $T = 100$ and $T = 250$. Figure A.3 shows histograms of the 5000 simulated values of $\hat{N}$, while Figure A.4 depicts histograms of the number of classification errors produced by our algorithm. By the number of classification errors, we simply mean the number of incorrectly classified time series, which is formally calculated as $\min_{\pi \in \Pi} \{|G_1 \setminus \hat{G}_{\pi(1)}| + \dots + |G_N \setminus \hat{G}_{\pi(N)}|\}$, where Π is the set of all permutations of $\{1, 2, \dots, \max\{N, \hat{N}\}\}$ and we let $\hat{G}_l = \emptyset$ for $\hat{N} < l \leq N$.

Table A.5: Clustering results for different time series lengths T and nominal sizes α .

(a) Empirical probabilities that $\hat{N} = N$

T	nominal size α		
	0.01	0.05	0.1
100	0.000	0.002	0.005
250	0.004	0.023	0.051
500	0.758	0.953	0.964

(b) Empirical probabilities that $\{\hat{G}_1, \dots, \hat{G}_{\hat{N}}\} = \{G_1, G_2, G_3\}$

T	nominal size α		
	0.01	0.05	0.1
100	0.000	0.000	0.000
250	0.001	0.008	0.020
500	0.751	0.947	0.958

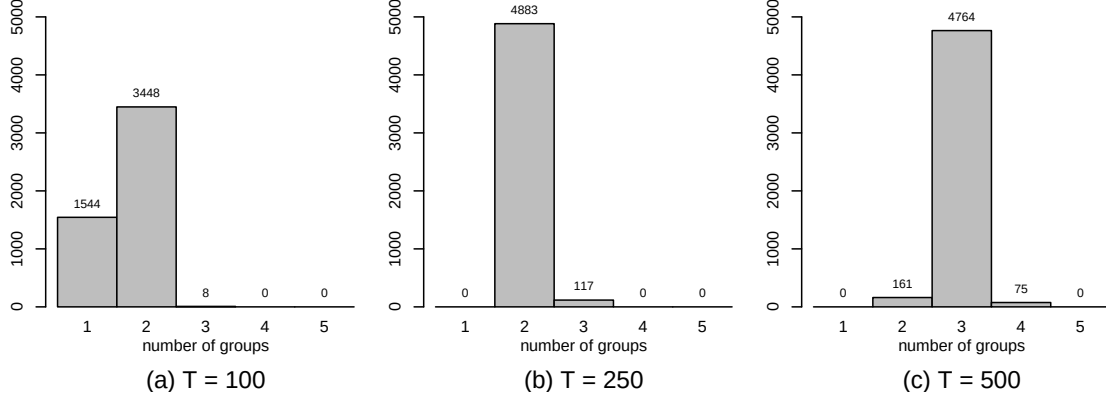


Figure A.3: Histograms of the estimated number of groups $\hat{N}$ for nominal size $\alpha = 0.05$.

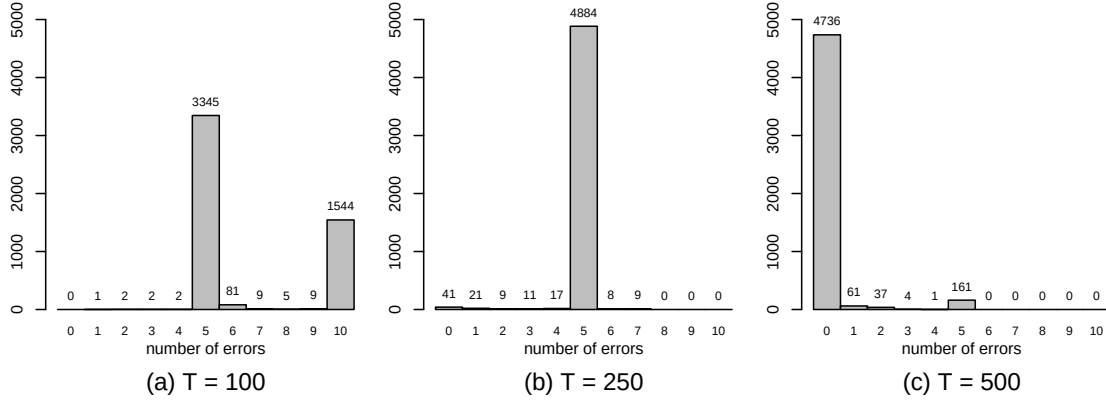


Figure A.4: Histograms of the number of classification errors for nominal size $\alpha = 0.05$.

The histograms of Figure A.3 for $T = 100$ and $T = 250$ clearly show that our method underestimates the number of groups. In particular, it fails to detect the difference between two out of three groups in a large number of simulations ($\hat{N} = 2$ in 3448 and 4883 cases out of 5000 for $T = 100$ and $T = 250$, respectively). This is reflected in the corresponding histograms of Figure A.4 which show that there are exactly 5 classification errors in 3345

(4884) of the 5000 simulation runs for $T = 100$ ($T = 250$). In most of these cases, the estimated group structure $\{\widehat{G}_1, \widehat{G}_2\}$ coincides with either $\{G_1 \cup G_2, G_3\}$, $\{G_1, G_2 \cup G_3\}$ or $\{G_1 \cup G_3, G_2\}$. In summary, we can conclude that the very small empirical probabilities for $T = 100$ and $T = 250$ in Table A.5 are due to the algorithm underestimating the number of groups. Inspecting the histograms for $T = 500$ in Figures A.3 and A.4, one can see that the performance of the estimators $\widehat{N}$ and $\{\widehat{G}_1, \dots, \widehat{G}_{\widehat{N}}\}$ improves significantly as the time series length T increases.

A.6 Comparison of our clustering algorithm with a benchmark

We finally compare our clustering algorithm with a simple benchmark. To do so, we consider a simplified version of the simulation setup described in Section A.5. In particular, we use the same setup but without covariates and fixed effects (i.e., $\beta_i = 0$ and $\alpha_i = 0$ for all i). The benchmark method proceeds as follows:

Step 1. For each i , estimate the trend function m_i by a standard local linear estimator

$$\widehat{m}_i = \widehat{m}_{i,h} \text{ with bandwidth } h.$$

Step 2. Measure the distance between $\widehat{m}_i$ and $\widehat{m}_j$ by the supremum metric

$$d_{ij}^{\text{bmk}} = \sup_{u \in [0,1]} |\widehat{m}_i(u) - \widehat{m}_j(u)|.$$

Step 3. Construct the following dissimilarity measure from these supremum distances:

$$\widehat{\Delta}^{\text{bmk}}(S, S') = \max_{i \in S, j \in S'} d_{ij}^{\text{bmk}}.$$

Step 4. Run a HAC algorithm with the computed dissimilarities.

Our multiscale approach can be regarded as a refinement of this very natural and intuitive benchmark method: it replaces the simple bandwidth-dependent distance measure d_{ij}^{bmk} by the multiscale distance measure $d_{ij} = \max_{(u,h) \in \mathcal{G}_T} \widehat{\psi}_{ij,T}^0(u, h)$ and provides a way to estimate the number of clusters N . As the benchmark algorithm does not come with an estimate of the number of clusters N , we assume $N = 3$ to be known in what follows.

Table A.6: Empirical probabilities that the estimated group structure is identical to the true group structure $\{G_1, G_2, G_3\}$. $\mathcal{T}_{\text{MS}}$ denotes our multiscale method and $\mathcal{T}_{\text{bmk}, l}$ the benchmark procedure with bandwidth h_l ($l = 1, \dots, 6$).

	$\mathcal{T}_{\text{MS}}$	$\mathcal{T}_{\text{bmk}, 1}$	$\mathcal{T}_{\text{bmk}, 2}$	$\mathcal{T}_{\text{bmk}, 3}$	$\mathcal{T}_{\text{bmk}, 4}$	$\mathcal{T}_{\text{bmk}, 5}$	$\mathcal{T}_{\text{bmk}, 6}$
$T = 100$	0.065	0.046	0.017	0.011	0.004	0.002	0.003
$T = 250$	0.476	0.439	0.222	0.056	0.018	0.006	0.006
$T = 500$	0.993	0.572	0.603	0.255	0.090	0.035	0.025

We implement our clustering algorithm in exactly the same way as in Section A.5 (with $N = 3$ known). To run the benchmark algorithm in practice, we need to discretize the supremum over all time points $u \in [0, 1]$ in Step 2 of the algorithm. To make the comparison as fair as possible, we replace this supremum by the maximum over all points $u \in U_T$, where U_T is the grid of locations defined in (A.2) that underlies the location-scale grid $\mathcal{G}_T = U_T \times H_T$ of our multiscale test. Moreover, we need to choose a bandwidth to run the benchmark algorithm. We consider six different choices: $h_l = h_{\min} + c_l(h_{\max} - h_{\min})$ with $c_l = 0.2(l - 1)$ for $l = 1, \dots, 6$.

The simulation results are reported in Table A.6. The entries of the table give the empirical probabilities with which the estimated group structure equals the true one $\{G_1, G_2, G_3\}$. The two main results are as follows: (i) The performance of the benchmark procedure is heavily influenced by the choice of bandwidth. (ii) Our multiscale method outperforms the benchmark with all considered bandwidth choices h_l . Results (i) and (ii) nicely illustrate the main advantage of our multiscale method: it automatically selects the “right” scale(s) or bandwidth(s), i.e., those which are most suitable for comparing the trend curves.

B Application to GDP growth data

In what follows, we revisit an application example from Zhang et al. (2012). The aim is to test the hypothesis of a common trend in the GDP growth time series of several OECD countries. Since we do not have access to the original dataset of Zhang et al. (2012) and

we do not know the exact data specifications used there, we work with data from the following common sources: Refinitiv Datastream, the OECD.Stat database, Federal Reserve Economics Data (FRED) and the Barro-Lee Educational Attainment dataset (Barro and Lee, 2013). We consider a data specification that is as close as possible to the one in Zhang et al. (2012) with one important distinction: In the original study, the authors examined 16 OECD countries (not specifying which ones) over the time period from the fourth quarter of 1975 up to and including the third quarter of 2010, whereas we consider only 11 countries (Australia, Austria, Canada, Finland, France, Germany, Japan, Norway, Switzerland, UK and USA) over the same time span. The reason is that we have access to data of good quality only for these 11 countries. In the following list, we specify the data for our analysis.⁵

- **Gross domestic product (*GDP*):** We use freely available data on *Gross Domestic Product – Expenditure Approach* from the OECD.Stat database (<https://stats.oecd.org/Index.aspx>). To be as close as possible to the specification of the data in Zhang et al. (2012), we use seasonally adjusted quarterly data on GDP expressed in millions of 2015 US dollars.⁶
- **Capital (*K*):** We use data on *Gross Fixed Capital Formation* from the OECD.Stat database. The data are at a quarterly frequency, seasonally adjusted, and expressed in millions of 2015 US dollars. In contrast to Zhang et al. (2012), who use data on *Capital Stock at Constant National Prices*, we choose to work with gross fixed capital formation due to data availability. It is worth noting that since accurate data on capital stock is notoriously difficult to collect, the use of gross fixed capital formation as a measure of capital is standard in the literature; see e.g. Sharma and Dhakal (1994), Lee and Huang (2002) and Lee (2005).

⁵All data were accessed and downloaded on 7 December 2021.

⁶Since the publication of Zhang et al. (2012), the OECD reference year has changed from 2005 to 2015. We have decided to analyse the latest version of the data in order to be able to make more accurate and up-to-date conclusions.

- **Labour (L):** We collect data on the *Number of Employed People* from various sources. For most of the countries (Austria, Australia, Canada, Germany, Japan, UK and USA), we download the OECD data on *Employed Population: Aged 15 and Over* retrieved from FRED (<https://fred.stlouisfed.org/>). The data for France and Switzerland were downloaded from Refinitiv Datastream. For all of the aforementioned countries, the observations are at a quarterly frequency and seasonally adjusted. The data for Finland and Norway were also obtained via Refinitiv Datastream, however, the only quarterly time series that are long enough for our purposes are not seasonally adjusted. Hence, for these two countries, we perform the seasonal adjustment ourselves. We in particular use the default method of the function `seas` from the R package `seasonal` (Sax and Eddelbuettel, 2018) which is an interface to X-13-ARIMA-SEATS, the seasonal adjustment software used by the US Census Bureau. For all of the countries, the data are given in thousands of persons.
- **Human capital (H):** We use *Educational Attainment for Population Aged 25 and Over* collected from <http://www.barrolee.com> as a measure of human capital. Since the only available data is five-year census data, we follow Zhang et al. (2012) and use linear interpolation between the observations and constant extrapolation on the boundaries (second and third quarters of 2010) to obtain the quarterly time series.

For each of the $n = 11$ countries in our sample, we thus observe a quarterly time series $\mathcal{T}_i = \{(Y_{it}, \mathbf{X}_{it}) : 1 \leq t \leq T\}$ of length $T = 140$, where $Y_{it} = \Delta \ln GDP_{it} := \ln GDP_{it} - \ln GDP_{i(t-1)}$ and $\mathbf{X}_{it} = (\Delta \ln L_{it}, \Delta \ln K_{it}, \Delta \ln H_{it})^\top$ with $\Delta \ln L_{it} := \ln L_{it} - \ln L_{i(t-1)}$, $\Delta \ln K_{it} := \ln K_{it} - \ln K_{i(t-1)}$ and $\Delta \ln H_{it} := \ln H_{it} - \ln H_{i(t-1)}$. Without loss of generality, we let $\Delta \ln GDP_{i1} = \Delta \ln L_{i1} = \Delta \ln K_{i1} = \Delta \ln H_{i1} = 0$. Each time series $\mathcal{T}_i$ is assumed to follow the model $Y_{it} = m_i(t/T) + \beta_i^\top \mathbf{X}_{it} + \alpha_i + \varepsilon_{it}$, or equivalently,

$$\Delta \ln GDP_{it} = m_i\left(\frac{t}{T}\right) + \beta_{i,1} \Delta \ln L_{it} + \beta_{i,2} \Delta \ln K_{it} + \beta_{i,3} \Delta \ln H_{it} + \alpha_i + \varepsilon_{it} \quad (\text{B.1})$$

for $1 \leq t \leq T$, where $\beta_i = (\beta_{i,1}, \beta_{i,2}, \beta_{i,3})^\top$ is a vector of unknown parameters, m_i is a country-specific nonparametric time trend and α_i is a country-specific fixed effect.

In order to test the null hypothesis $H_0 : m_1 = \dots = m_n$ with $n = 11$ in model (B.1), we implement our multiscale test as follows:

- We choose K to be the Epanechnikov kernel and consider the set of location-scale points $\mathcal{G}_T = U_T \times H_T$, where $U_T = \{u \in [0, 1] : u = \frac{8t+1}{2T} \text{ for some } t \in \mathbb{N}\}$ and $H_T = \{h \in [\frac{\log T}{T}, \frac{1}{4}] : h = \frac{4t}{T} \text{ for some } t \in \mathbb{N}\}$. We thus take into account all locations u on an equidistant grid U_T with step length $4/T$ and all scales $h = 4/T, 8/T, 12/T, \dots$ with $\log T/T \leq h \leq 1/4$. The choice of the grid $\mathcal{G}_T$ is motivated by the quarterly frequency of the data: each interval $\mathcal{I}_{u,h}$ with $(u, h) \in \mathcal{G}_T$ spans 8, 16, 24, $\dots$ quarters, i.e., 2, 4, 6, $\dots$ years. The lower bound $\log T/T$ on the scales h in H_T is motivated by Assumption (C11).
- To obtain an estimator $\hat{\sigma}_i^2$ of the long-run error variance σ_i^2 for each i , we assume that the error process $\mathcal{E}_i$ follows an $\text{AR}(p_i)$ model and apply the difference-based procedure of Khismatullina and Vogt (2020) to the augmented time series $\{\hat{Y}_{it} : 1 \leq t \leq T\}$ with $\hat{Y}_{it} = Y_{it} - \hat{\beta}_i^\top \mathbf{X}_{it} - \hat{\alpha}_i$. We set the tuning parameters q and r of the procedure to 20 and 10, respectively, and choose the AR order p_i by minimizing BIC, which yields $p_i = 3$ for Australia, Canada and the UK and $p_i = 1$ for all other countries.⁷
- The critical values $q_{n,T}(\alpha)$ are computed by Monte Carlo methods as described in Section 3.4, where we set $L = 5000$.

Besides these choices, we construct and implement the multiscale test exactly as described in Section 3.

The thus implemented multiscale test rejects the global null hypothesis H_0 at the usual significance levels $\alpha = 0.01, 0.05, 0.1$. This result is in line with the findings in Zhang et al. (2012) where the null hypothesis of a common trend is rejected at level $\alpha = 0.1$. The

⁷We also calculated the values of other information criteria such as FPE, AIC and HQ which, in most of the cases, resulted in the same values of p_i .

main advantage of our multiscale test over the method in Zhang et al. (2012) is that it is much more informative. In particular, it provides information about *which* of the $n = 11$ countries have different trends and *where* the trends differ. This information is presented graphically in Figures B.1–B.11. Each figure corresponds to a specific pair of countries (i, j) and is divided into three panels (a)–(c). Panel (a) shows the augmented time series $\{\hat{Y}_{it} : 1 \leq t \leq T\}$ and $\{\hat{Y}_{jt} : 1 \leq t \leq T\}$ for the two countries i and j that are compared. Panel (b) presents smoothed versions of the time series from (a), in particular, it shows local linear kernel estimates of the two trend functions m_i and m_j , where the bandwidth is set to 14 quarters (that is, to 0.1 in terms of rescaled time) and an Epanechnikov kernel is used. Panel (c) presents the results produced by our test for the significance level $\alpha = 0.05$: it depicts in grey the set $\mathcal{S}^{[i,j]}(\alpha)$ of all the intervals for which the test rejects the local null $H_0^{[i,j]}(u, h)$. The set of minimal intervals $\mathcal{S}_{\min}^{[i,j]}(\alpha) \subseteq \mathcal{S}^{[i,j]}(\alpha)$ is highlighted in black. According to (4.3), we can make the following simultaneous confidence statement about the intervals plotted in panels (c) of Figures B.1–B.11: we can claim, with confidence of about 95%, that there is a difference between the functions m_i and m_j on each of these intervals.

Out of 55 pairwise comparisons, our test detects differences for 11 pairs of countries (i, j) . These 11 cases are presented in Figures B.1–B.11. In 9 cases (Figures B.1–B.9), one of the involved countries is Norway. Inspecting the trend estimates in panels (b) of Figures B.1–B.9, the Norwegian trend estimate can be seen to exhibit a strong downward movement at the end of the observation period, whereas the other trend estimates show a much less pronounced downward movement (or even a slight upward movement). According to our test, this is a significant difference between the Norwegian and the other trend functions rather than an artefact of the sampling noise: In all 9 cases, the test rejects the local null for at least one interval which covers the last 10 years of the analysed time period (from the first quarter in 2000 up to the third quarter in 2010). Apart from these differences at

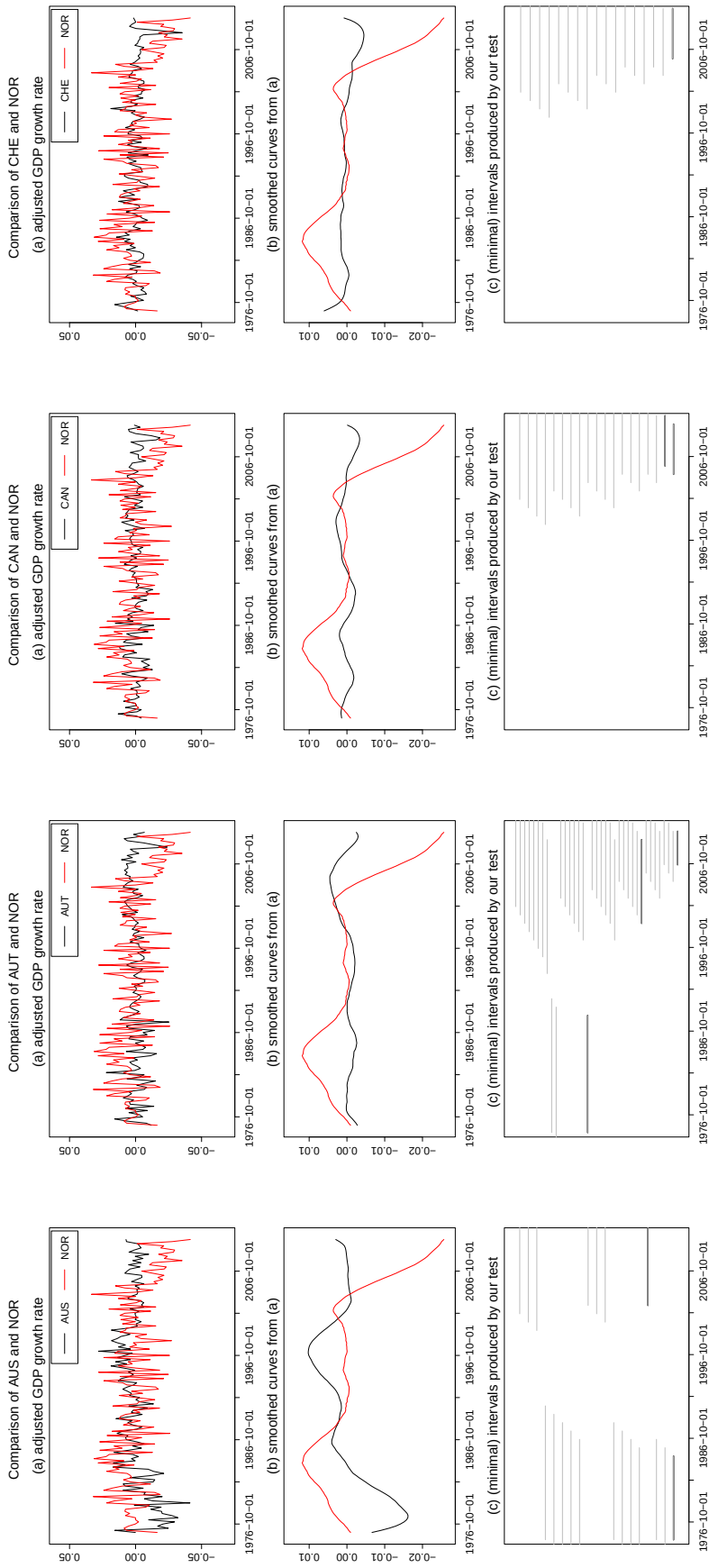


Figure B.1: Test results for the comparison of Australia and Norway. Figure B.2: Test results for the comparison of Austria and Norway. Figure B.3: Test results for the comparison of Canada and Norway. Figure B.4: Test results for the comparison of Switzerland and Norway.

Note: In each figure, panel (a) shows the two augmented time series, panel (b) presents smoothed versions of the augmented time series, and panel (c) depicts the set of intervals $\mathcal{S}^{[i,j]}(\alpha)$ in grey and the subset of minimal intervals $\mathcal{S}_{\min}^{[i,j]}(\alpha)$ in black.

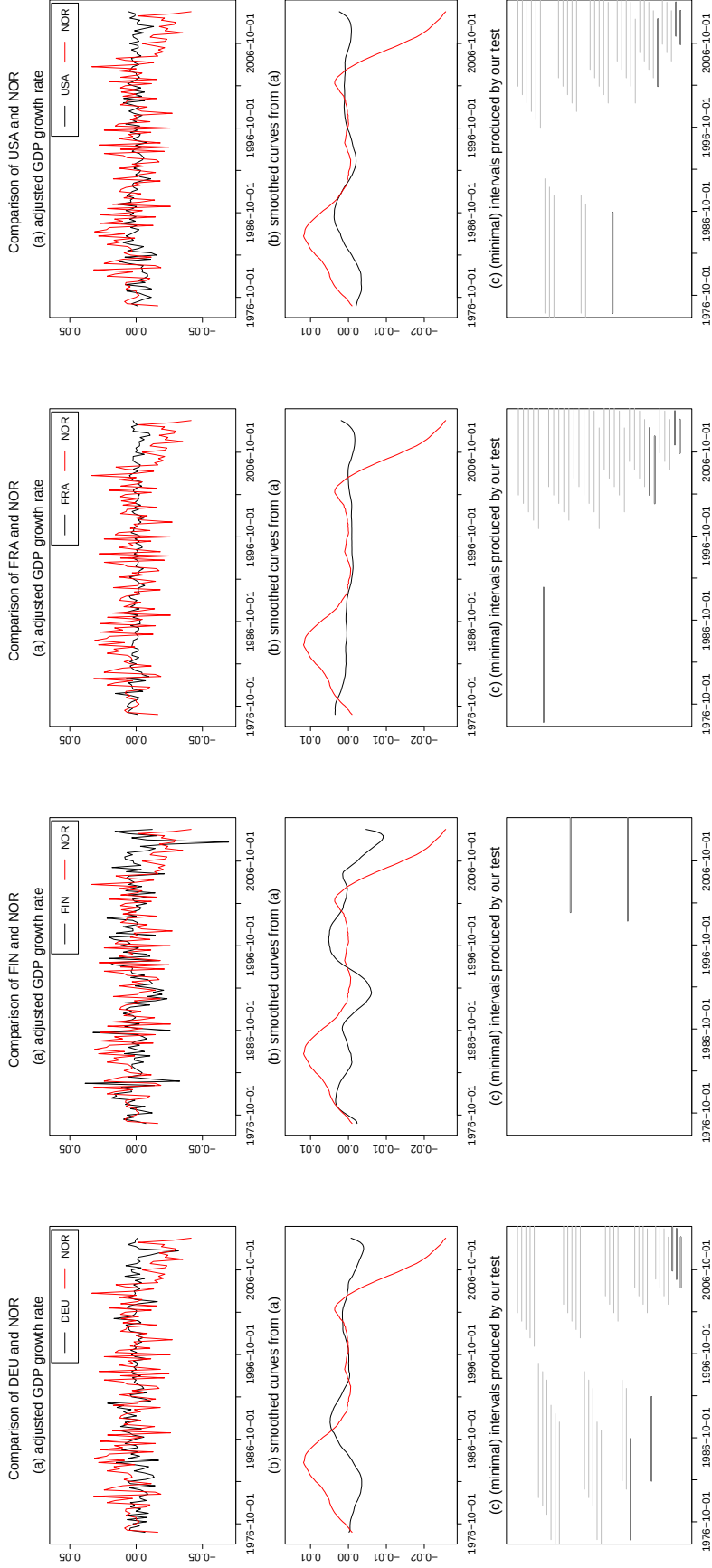


Figure B.5: Test results for the comparison of Germany and Norway. way. Figure B.6: Test results for the comparison of Finland and Norway. way. Figure B.7: Test results for the comparison of France and Norway. way. Figure B.8: Test results for the comparison of the USA and Norway. way.

Note: In each figure, panel (a) shows the two augmented time series, panel (b) presents smoothed versions of the augmented time series, and panel (c) depicts the set of intervals $\mathcal{S}^{[i,j]}(\alpha)$ in grey and the subset of minimal intervals $\mathcal{S}_{\min}^{[i,j]}(\alpha)$ in black.

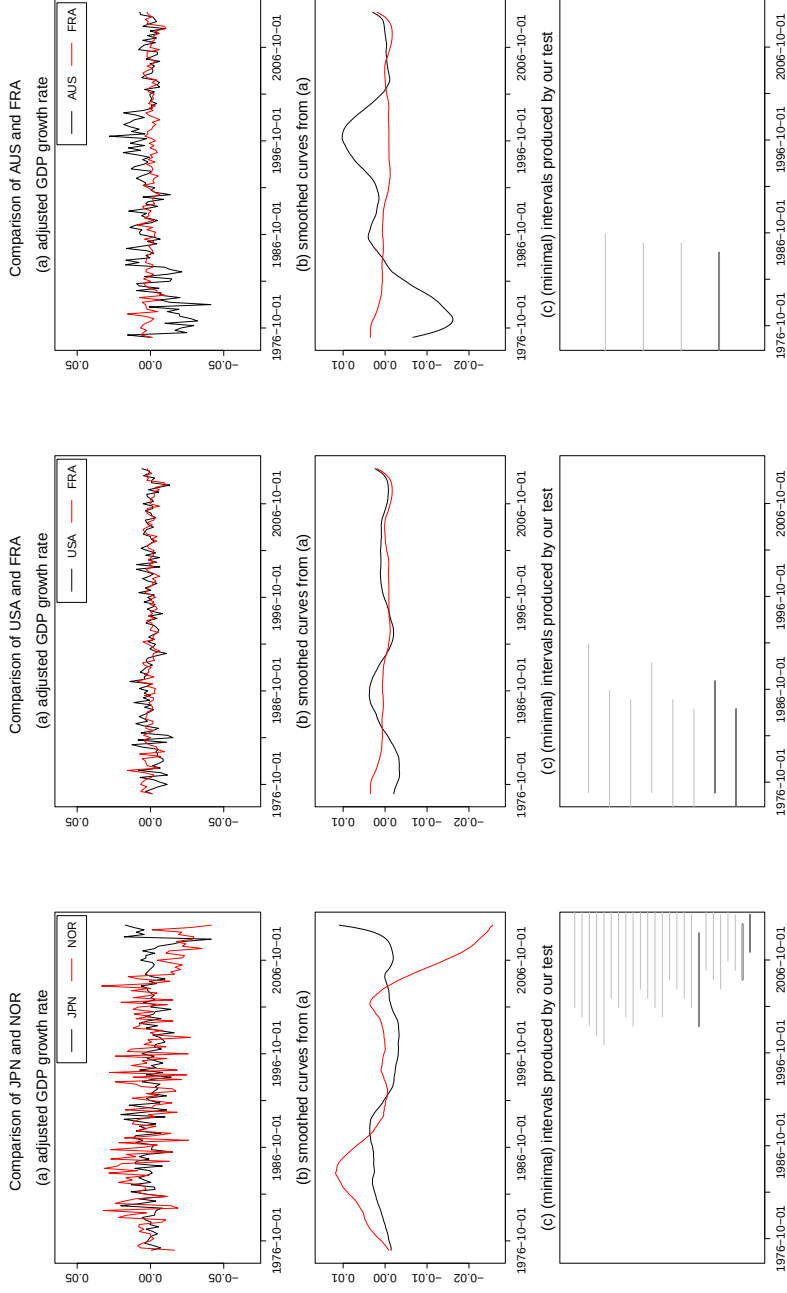


Figure B.9: Test results for the comparison of Japan and Norway. Figure B.10: Test results for the comparison of the USA and France. Figure B.11: Test results for the comparison of Australia and France.

Note: In each figure, panel (a) shows the two augmented time series, panel (b) presents smoothed versions of the augmented time series, and panel (c) depicts the set of intervals $\mathcal{S}^{[i,j]}(\alpha)$ in grey and the subset of minimal intervals $\mathcal{S}_{\min}^{[i,j]}(\alpha)$ in black.

the end of the sampling period, our test also finds differences in the beginning, however, only for part of the pairwise comparisons.

Figures B.10 and B.11 present the results of the pairwise comparison between Australia and France and between the USA and France, respectively. In both cases, our test detects differences between the GDP trends only in the beginning of the considered time period. In the case of Australia and France, it is clearly visible in the raw data (panel (a) in Figure B.11) that there is a difference between the trends, whereas this is not so obvious in the case of the USA and France. According to our test, there are indeed significant differences in both cases. In particular, we can claim with confidence at least 95%, that there are differences between the trends of the USA and France (of Australia and France) up to the fourth quarter in 1991 (the fourth quarter in 1986), but there is no evidence of any differences between the trends from 1992 (1987) onwards.

We next apply our clustering techniques to find groups of countries that have the same time trend. We implement our HAC algorithm with $\alpha = 0.05$ and the same choices as detailed above. The dendrogram that depicts the clustering results is plotted in Figure B.12. The number of clusters is estimated to be $\hat{N} = 3$. The rectangles in Figure B.12 indicate the $\hat{N} = 3$ clusters. In particular, each rectangle is drawn around the branches of the dendrogram that correspond to one of the three clusters. Figure B.13 depicts local linear kernel estimates of the $n = 11$ GDP time trends (calculated from the augmented time series $\hat{Y}_{it}$ with bandwidth 0.1 and Epanechnikov kernel). Their colour indicates which cluster they belong to.

The results in Figures B.12 and B.13 show that there is one cluster which consists only of Norway (plotted in red). As we have already discussed above and as becomes apparent from Figure B.13, the Norwegian trend exhibits a strong downward movement at the end of the sampling period, whereas the other trends show a much more moderate downward movement (if at all). This is presumably the reason why the clustering procedure puts

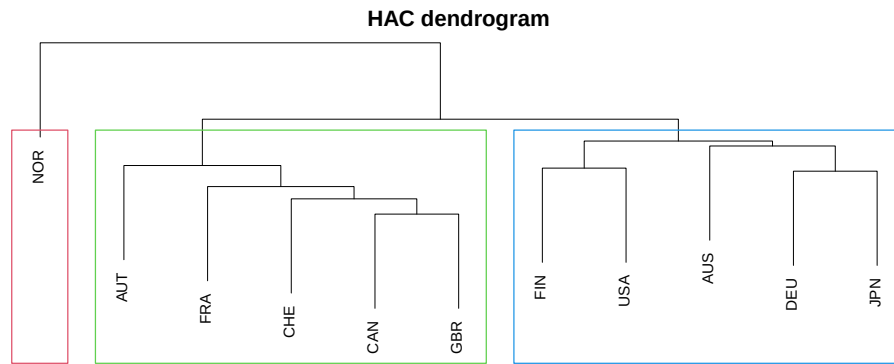


Figure B.12: Dendrogram of the HAC algorithm. Each coloured rectangle corresponds to one of the clusters.

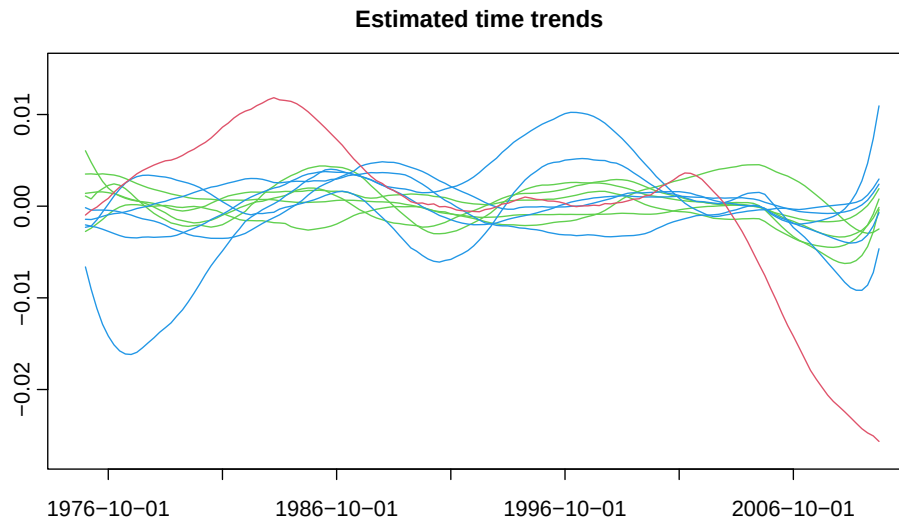


Figure B.13: Local linear estimates of the $n = 11$ time trends (calculated from the augmented time series $\hat{Y}_{it}$ with bandwidth $h = 0.1$ and Epanechnikov kernel). Each trend estimate is coloured according to the cluster that it is assigned to.

Norway in a separate cluster. The algorithm further finds two other clusters, one consisting of the 5 countries Australia, Finland, Germany, Japan and the USA (plotted in blue in Figures B.12 and B.13) and the other one consisting of the 5 countries Austria, Canada, France, Switzerland and the UK (plotted in green in Figures B.12 and B.13). Visual inspection of the trend estimates in Figure B.13 suggests that the GDP time trends in the blue cluster exhibit more pronounced decreases and increases than the GDP time trends in the green cluster. Hence, overall, the clustering procedure appears to produce a reasonable grouping of the GDP trends.

C Technical details

In what follows, we prove the theoretical results from Sections 4 and 5. We use the following notation: The symbol C denotes a universal real constant which may take a different value on each occurrence. For $a, b \in \mathbb{R}$, we write $a \vee b = \max\{a, b\}$. For $x \in \mathbb{R}_{\geq 0}$, we let $\lfloor x \rfloor$ denote the integer value of x and $\lceil x \rceil$ the smallest integer greater than or equal to x . For any set A , the symbol $|A|$ denotes the cardinality of A . The expression $X \stackrel{\mathcal{D}}{=} Y$ means that the two random variables X and Y have the same distribution. Finally, we sometimes use the notation $a_T \ll b_T$ to express that $a_T = o(b_T)$.

Auxiliary results

Let $\{Z_t\}_{t=-\infty}^{\infty}$ be a stationary time series process with $Z_t \in \mathcal{L}^q$ for some $q > 2$ and $\mathbb{E}[Z_t] = 0$. Assume that Z_t can be represented as $Z_t = g(\dots, \eta_{t-1}, \eta_t)$, where η_t are i.i.d. variables and $g : \mathbb{R}^{\infty} \rightarrow \mathbb{R}$ is a measurable function. We first state a Nagaev-type inequality from Wu and Wu (2016).

Definition C.1. Let $q > 0$ and $\alpha > 0$. The dependence adjusted norm of the process $Z. = \{Z_t\}_{t=-\infty}^{\infty}$ is given by $\|Z.\|_{q,\alpha} = \sup_{t \geq 0} (t+1)^{\alpha} \sum_{s=t}^{\infty} \delta_q(g, s)$.

Proposition C.2 (Wu and Wu (2016), Theorem 2). Assume that $\|Z.\|_{q,\alpha} < \infty$ with $q > 2$ and $\alpha > 1/2 - 1/q$. Let $S_T = a_1 Z_1 + \dots + a_T Z_T$, where $a_1, \dots, a_T$ are real numbers with $\sum_{t=1}^T a_t^2 = T$. Then for any $w > 0$,

$$\mathbb{P}(|S_T| \geq w) \leq C_1 \frac{|a|_q^q \|Z.\|_{q,\alpha}^q}{w^q} + C_2 \exp\left(-\frac{C_3 w^2}{T \|Z.\|_{2,\alpha}^2}\right),$$

where C_1, C_2, C_3 are constants that only depend on q and α .

The following lemma is a simple consequence of the above inequality.

Lemma C.3. Let $\sum_{s=t}^{\infty} \delta_q(g, s) = O(t^{-\alpha})$ for some $q > 2$ and $\alpha > 1/2 - 1/q$. Then

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T Z_t = O_p(1).$$

Proof of Lemma C.3. Let $\eta > 0$ be a fixed number. We apply Proposition C.2 to the sum $S_T = \sum_{t=1}^T a_t Z_t$ with $a_t = 1$ for all t . The assumption $\sum_{s=t}^{\infty} \delta_q(g, s) = O(t^{-\alpha})$ implies that $\|Z\|_{2,\alpha} \leq \|Z\|_{q,\alpha} \leq C_Z < \infty$. Hence, for w chosen sufficiently large, we get

$$\begin{aligned} \mathbb{P}\left(\left|\sum_{t=1}^T Z_t\right| \geq \sqrt{T}w\right) &\leq C_1 \frac{TC_Z^q}{T^{q/2}w^q} + C_2 \exp\left(-\frac{C_3Tw^2}{TC_Z^2}\right) \\ &= \frac{\{C_1C_Z^q\}T^{1-q/2}}{w^q} + C_2 \exp\left(-\frac{C_3w^2}{C_Z^2}\right) \leq \eta \end{aligned}$$

for all T . This means that $\sum_{t=1}^T Z_t/\sqrt{T} = O_p(1)$. $\square$

Let $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{it-1}$ and $\Delta\mathbf{X}_{it} = \mathbf{X}_{it} - \mathbf{X}_{it-1}$. By Assumptions (C1) and (C4), $\Delta\varepsilon_{it} = \Delta g_i(\mathcal{F}_{it})$ and $\Delta\mathbf{X}_{it} = \Delta\mathbf{h}_i(\mathcal{G}_{it})$. We further define

$$\begin{aligned} \mathbf{a}_i(\mathcal{H}_{it}) &:= \Delta\mathbf{h}_i(\mathcal{G}_{it})\Delta g_i(\mathcal{F}_{it}) = \Delta\mathbf{X}_{it}\Delta\varepsilon_{it} \\ \mathbf{b}_i(\mathcal{G}_{it}) &:= \Delta\mathbf{h}_i(\mathcal{G}_{it})\Delta\mathbf{h}_i(\mathcal{G}_{it})^\top = \Delta\mathbf{X}_{it}\Delta\mathbf{X}_{it}^\top, \end{aligned}$$

where $\mathbf{a}_i = (a_{ij})_{j=1}^d$, $\mathbf{b}_i = (b_{ikl})_{k,l=1}^d$ and $\mathcal{H}_{it} = (\mathcal{H}_{it,1}, \dots, \mathcal{H}_{it,d})^\top$ with $\mathcal{H}_{it,j} = (\dots, \nu_{it-1,j}, \nu_{it,j})$ and $\nu_{it,j} = (\eta_{it}, \xi_{it,j})$. The next result gives bounds on the physical dependence measures of the processes $\{\mathbf{a}_i(\mathcal{H}_{it})\}_{t=-\infty}^{\infty}$ and $\{\mathbf{b}_i(\mathcal{G}_{it})\}_{t=-\infty}^{\infty}$.

Lemma C.4. *Let Assumptions (C1), (C3), (C4) and (C6) be satisfied. Then for each i , j , k and l , it holds that*

$$\begin{aligned} \sum_{s=t}^{\infty} \delta_p(a_{ij}, s) &= O(t^{-\alpha}) \quad \text{for } p = \min\{q, q'\}/2 \text{ and some } \alpha > 1/2 - 1/p \\ \sum_{s=t}^{\infty} \delta_p(b_{ikl}, s) &= O(t^{-\alpha}) \quad \text{for } p = q'/2 \text{ and some } \alpha > 1/2 - 1/p. \end{aligned}$$

Proof of Lemma C.4. We only prove the first statement. The second one follows by analogous arguments. By the definition of the physical dependence measure and the Cauchy-Schwarz inequality, we have with $p = \min\{q, q'\}/2$ that

$$\begin{aligned} \delta_p(a_{ij}, s) &= \|a_{ij}(\mathcal{H}_{it,j}) - a_{ij}(\mathcal{H}'_{it,j})\|_p \\ &= \|\Delta h_{ij}(\mathcal{G}_{it})\Delta g_i(\mathcal{F}_{it}) - \Delta h_{ij}(\mathcal{G}'_{it})\Delta g_i(\mathcal{F}'_{it})\|_p \end{aligned}$$

$$\begin{aligned}
&\leq \|h_{ij}(\mathcal{G}_{it})g_i(\mathcal{F}_{it}) - h_{ij}(\mathcal{G}'_{it})g_i(\mathcal{F}'_{it})\|_p \\
&\quad + \|h_{ij}(\mathcal{G}_{it-1})g_i(\mathcal{F}_{it-1}) - h_{ij}(\mathcal{G}'_{it-1})g_i(\mathcal{F}'_{it-1})\|_p \\
&\quad + \|h_{ij}(\mathcal{G}_{it-1})g_i(\mathcal{F}_{it}) - h_{ij}(\mathcal{G}'_{it-1})g_i(\mathcal{F}'_{it})\|_p \\
&\quad + \|h_{ij}(\mathcal{G}_{it})g_i(\mathcal{F}_{it-1}) - h_{ij}(\mathcal{G}'_{it})g_i(\mathcal{F}'_{it-1})\|_p \\
&= \|\{h_{ij}(\mathcal{G}_{it}) - h_{ij}(\mathcal{G}'_{it})\}g_i(\mathcal{F}_{it}) + h_{ij}(\mathcal{G}'_{it})\{g_i(\mathcal{F}_{it}) - g_i(\mathcal{F}'_{it})\}\|_p \\
&\quad + \|\{h_{ij}(\mathcal{G}_{it-1}) - h_{ij}(\mathcal{G}'_{it-1})\}g_i(\mathcal{F}_{it-1}) + h_{ij}(\mathcal{G}'_{it-1})\{g_i(\mathcal{F}_{it-1}) - g_i(\mathcal{F}'_{it-1})\}\|_p \\
&\quad + \|\{h_{ij}(\mathcal{G}_{it-1}) - h_{ij}(\mathcal{G}'_{it-1})\}g_i(\mathcal{F}_{it}) + h_{ij}(\mathcal{G}'_{it-1})\{g_i(\mathcal{F}_{it}) - g_i(\mathcal{F}'_{it})\}\|_p \\
&\quad + \|\{h_{ij}(\mathcal{G}_{it}) - h_{ij}(\mathcal{G}'_{it})\}g_i(\mathcal{F}_{it-1}) + h_{ij}(\mathcal{G}'_{it})\{g_i(\mathcal{F}_{it-1}) - g_i(\mathcal{F}'_{it-1})\}\|_p \\
&\leq \delta_{2p}(h_{ij}, t)\|g_i(\mathcal{F}_t)\|_{2p} + \delta_{2p}(g_i, t)\|h_{ij}(\mathcal{G}'_{it})\|_{2p} \\
&\quad + \delta_{2p}(h_{ij}, t-1)\|g_i(\mathcal{F}_{t-1})\|_{2p} + \delta_{2p}(g_i, t-1)\|h_{ij}(\mathcal{G}'_{it-1})\|_{2p} \\
&\quad + \delta_{2p}(h_{ij}, t-1)\|g_i(\mathcal{F}_t)\|_{2p} + \delta_{2p}(g_i, t)\|h_{ij}(\mathcal{G}'_{it-1})\|_{2p} \\
&\quad + \delta_{2p}(h_{ij}, t)\|g_i(\mathcal{F}_{t-1})\|_{2p} + \delta_{2p}(g_i, t-1)\|h_{ij}(\mathcal{G}'_{it})\|_{2p},
\end{aligned}$$

where $\mathcal{H}'_{it,j} = (\dots, \nu_{i(-1),j}, \nu'_{i0,j}, \nu_{i1,j}, \dots, \nu_{it-1,j}, \nu_{it,j})$, $\mathcal{G}'_{it,j} = (\dots, \xi_{i(-1),j}, \xi'_{i0,j}, \xi_{i1,j}, \dots, \xi_{it-1,j}, \xi_{it,j})$ and $\mathcal{F}'_{it} = (\dots, \eta_{i(-1)}, \eta'_{i0}, \eta_{i1}, \dots, \eta_{it-1}, \eta_{it})$ are coupled processes with $\nu'_{i0,j}$, $\xi'_{i0,j}$ and $\eta'_{i0,j}$ being i.i.d. copies of $\nu_{i0,j}$, $\xi_{i0,j}$ and η_{i0} . From this and Assumptions (C1), (C3), (C4) and (C6), it immediately follows that $\sum_{s=t}^{\infty} \delta_p(a_{ij}, s) = O(t^{-\alpha})$. $\square$

We now show that the estimator $\widehat{\beta}_i$ is $\sqrt{T}$ -consistent for each i under our conditions.

Lemma C.5. *Let Assumptions (C1), (C3) and (C4)–(C7) be satisfied. Then for each i ,*

$$\widehat{\beta}_i - \beta_i = O_p\left(\frac{1}{\sqrt{T}}\right).$$

Proof of Lemma C.5. The estimator $\widehat{\beta}_i$ can be written as

$$\begin{aligned}
\widehat{\beta}_i &= \left(\sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta Y_{it} \\
&= \left(\sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \sum_{t=2}^T \Delta \mathbf{X}_{it} \left(\Delta \mathbf{X}_{it}^\top \beta_i + \Delta m_{it} + \Delta \varepsilon_{it} \right)
\end{aligned}$$

$$= \beta_i + \left(\sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta m_{it} + \left(\sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \varepsilon_{it},$$

where $\Delta \mathbf{X}_{it} = \mathbf{X}_{it} - \mathbf{X}_{it-1}$, $\Delta \varepsilon_{it} = \varepsilon_{it} - \varepsilon_{it-1}$ and $\Delta m_{it} = m_i(\frac{t}{T}) - m_i(\frac{t-1}{T})$. Hence,

$$\begin{aligned} \sqrt{T}(\hat{\beta}_i - \beta_i) &= \left(\frac{1}{T} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta m_{it} \\ &\quad + \left(\frac{1}{T} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} \frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \varepsilon_{it}. \end{aligned} \quad (\text{C.1})$$

In what follows, we show that

$$\frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \varepsilon_{it} = O_p(1) \quad (\text{C.2})$$

$$\left(\frac{1}{T} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top \right)^{-1} = O_p(1) \quad (\text{C.3})$$

$$\frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta \mathbf{X}_{it} \Delta m_{it} = O_p\left(\frac{1}{\sqrt{T}}\right). \quad (\text{C.4})$$

Lemma C.5 follows from applying these three statements together with standard arguments to formula (C.1).

Since $\mathbb{E}[\Delta \mathbf{X}_{it} \Delta \varepsilon_{it}] = 0$ by (C7) and $\sum_{s=t}^\infty \delta_p(a_{ij}, s) = O(t^{-\alpha})$ for some $p > 2$, $\alpha > 1/2 - 1/p$ and all j by Lemma C.4, the claim (C.2) follows upon applying Lemma C.3. Another application of Lemma C.3 yields that

$$\frac{1}{T} \sum_{t=2}^T \left\{ \Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top - \mathbb{E}[\Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top] \right\} = O_p\left(\frac{1}{\sqrt{T}}\right).$$

As $\mathbb{E}[\Delta \mathbf{X}_{it} \Delta \mathbf{X}_{it}^\top]$ is invertible, we can invoke Slutsky's lemma to obtain (C.3). By assumption, m_i is Lipschitz continuous, which implies that $|\Delta m_{it}| = |m_i(\frac{t}{T}) - m_i(\frac{t-1}{T})| \leq C/T$ for all $t \in \{1, \dots, T\}$ and some constant $C > 0$. Hence,

$$\begin{aligned} \left| \frac{1}{\sqrt{T}} \sum_{t=2}^T \Delta X_{it,j} \Delta m_{it} \right| &\leq \frac{1}{\sqrt{T}} \sum_{t=2}^T |\Delta X_{it,j}| \cdot |\Delta m_{it}| \\ &\leq \frac{C}{\sqrt{T}} \cdot \frac{1}{T} \sum_{t=2}^T |\Delta X_{it,j}| = O_p\left(\frac{1}{\sqrt{T}}\right), \end{aligned}$$

where we have used that $T^{-1} \sum_{t=2}^T |\Delta X_{it,j}| = O_p(1)$ by Markov's inequality. This yields (C.4). $\square$

Lemma C.6. *Let*

$$\hat{\sigma}_i^2 = \frac{1}{2(M-1)s_T} \sum_{m=1}^M \left[\sum_{t=1}^{s_T} \left(Y_{i(t+ms_T)} - Y_{i(t+(m-1)s_T)} - \hat{\beta}_i^\top (\mathbf{X}_{i(t+ms_T)} - \mathbf{X}_{i(t+(m-1)s_T)}) \right) \right]^2,$$

be the subseries estimator of σ_i^2 , where $s_T \asymp T^{1/3}$ is the length of the subseries and $M = \lfloor T/s_T \rfloor$ is the largest integer not exceeding T/s_T . Under Assumptions (C1)–(C7),

$$\hat{\sigma}_i^2 = \sigma_i^2 + O_p(T^{-1/3})$$

for each i .

Proof of Lemma C.6. Let $Y_{it}^\circ := m_i(t/T) + \varepsilon_{it}$. Using simple arithmetic calculations, we can rewrite $\hat{\sigma}_i^2$ as $\hat{\sigma}_i^2 = \hat{\sigma}_{i,A}^2 + \hat{\sigma}_{i,B}^2 - \hat{\sigma}_{i,C}^2$, where

$$\begin{aligned} \hat{\sigma}_{i,A}^2 &= \frac{1}{2(M-1)s_T} \sum_{m=1}^M \left[\sum_{t=1}^{s_T} (Y_{i(t+ms_T)}^\circ - Y_{i(t+(m-1)s_T)}^\circ) \right]^2 \\ \hat{\sigma}_{i,B}^2 &= \frac{1}{2(M-1)s_T} \sum_{m=1}^M \left[\sum_{t=1}^{s_T} (\hat{\beta}_i - \beta_i)^\top (\mathbf{X}_{i(t+ms_T)} - \mathbf{X}_{i(t+(m-1)s_T)}) \right]^2 \\ \hat{\sigma}_{i,C}^2 &= \frac{1}{(M-1)s_T} \sum_{m=1}^M \left[\sum_{t=1}^{s_T} (Y_{i(t+ms_T)}^\circ - Y_{i(t+(m-1)s_T)}^\circ) \right. \\ &\quad \left. \times \sum_{t=1}^{s_T} (\hat{\beta}_i - \beta_i)^\top (\mathbf{X}_{i(t+ms_T)} - \mathbf{X}_{i(t+(m-1)s_T)}) \right]. \end{aligned}$$

By Carlstein (1986) and Wu and Zhao (2007), $\hat{\sigma}_{i,A}^2 = \sigma_i^2 + O_p(T^{-1/3})$. Moreover, under our assumptions, it is straightforward to see that $\hat{\sigma}_{i,B}^2 = O_p(T^{-1/3})$ and $\hat{\sigma}_{i,C}^2 = O_p(T^{-1/3})$. $\square$

Proof of Theorem 4.1

We first summarize the main proof strategy, which splits up into five steps, and then fill in the details. We in particular defer the proofs of some intermediate results to the end of the section.

Step 1

To start with, we consider a simplified setting where the parameter vectors β_i are known. In this case, we can replace the estimators $\hat{\beta}_i$ in the definition of the statistic $\hat{\Phi}_{n,T}$ by the true vectors β_i themselves. This leads to the simpler statistic

$$\hat{\hat{\Phi}}_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\hat{\phi}_{ij,T}(u,h)}{\{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{1/2}} \right| - \lambda(h) \right\},$$

where

$$\hat{\phi}_{ij,T}(u,h) = \sum_{t=1}^T w_{t,T}(u,h) \{(\varepsilon_{it} - \bar{\varepsilon}_i) - (\varepsilon_{jt} - \bar{\varepsilon}_j)\}$$

and $\hat{\sigma}_i^2$ is computed in exactly the same way as $\hat{\sigma}_i^2$ except that all occurrences of $\hat{\beta}_i$ are replaced by β_i . By assumption, $\hat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$ with $\rho_T = o(1/\log T)$. For most estimators of σ_i^2 including those discussed in Section 4, this assumption immediately implies that $\hat{\hat{\sigma}}_i^2 = \sigma_i^2 + o_p(\rho_T)$ as well. In the sequel, we thus take for granted that the estimator $\hat{\hat{\sigma}}_i^2$ has this property.

We now have a closer look at the statistic $\hat{\hat{\Phi}}_{n,T}$. We in particular show that there exists an identically distributed version $\tilde{\Phi}_{n,T}$ of $\hat{\hat{\Phi}}_{n,T}$ which is close to the Gaussian statistic $\Phi_{n,T}$ from (3.10). More formally, we prove the following result.

Proposition C.7. *There exist statistics $\{\tilde{\Phi}_{n,T} : T = 1, 2, \dots\}$ with the following two properties: (i) $\tilde{\Phi}_{n,T}$ has the same distribution as $\hat{\hat{\Phi}}_{n,T}$ for any T , and (ii)*

$$|\tilde{\Phi}_{n,T} - \Phi_{n,T}| = o_p(\delta_T),$$

where $\delta_T = T^{1/q}/\sqrt{Th_{\min}} + \rho_T\sqrt{\log T}$ and $\Phi_{n,T}$ is a Gaussian statistic as defined in (3.10).

The proof makes heavy use of strong approximation theory for dependent processes. As it is quite technical, it is postponed to the end of this section.

Step 2

In this step, we establish some properties of the Gaussian statistic $\Phi_{n,T}$. Specifically, we prove the following result.

Proposition C.8. *It holds that*

$$\sup_{x \in \mathbb{R}} \mathbb{P}(|\Phi_{n,T} - x| \leq \delta_T) = o(1),$$

where $\delta_T = T^{1/q} / \sqrt{Th_{\min}} + \rho_T \sqrt{\log T}$.

Roughly speaking, this proposition says that the random variable $\Phi_{n,T}$ does not concentrate too strongly in small regions of the form $[x - \delta_T, x + \delta_T]$ with δ_T converging to 0. The main technical tool for deriving it are anti-concentration bounds for Gaussian random vectors. The details are provided below.

Step 3

We now use Steps 1 and 2 to prove that

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x)| = o(1). \quad (\text{C.5})$$

Proof of (C.5). It holds that

$$\begin{aligned} & \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| \\ &= \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widetilde{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| \\ &= \sup_{x \in \mathbb{R}} \left| \mathbb{E} \left[\mathbb{1}(\widetilde{\Phi}_{n,T} \leq x) - \mathbb{1}(\Phi_{n,T} \leq x) \right] \right| \\ &\leq \sup_{x \in \mathbb{R}} \left| \mathbb{E} \left[\left\{ \mathbb{1}(\widetilde{\Phi}_{n,T} \leq x) - \mathbb{1}(\Phi_{n,T} \leq x) \right\} \mathbb{1}(|\widetilde{\Phi}_{n,T} - \Phi_{n,T}| \leq \delta_T) \right] \right| \\ &\quad + \mathbb{E} \left[\mathbb{1}(|\widetilde{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T) \right]. \end{aligned}$$

Moreover, since

$$\mathbb{E} \left[\mathbb{1}(|\widetilde{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T) \right] = \mathbb{P}(|\widetilde{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T) = o(1)$$

by Step 1 and

$$\begin{aligned}
& \sup_{x \in \mathbb{R}} \left| \mathbb{E} \left[\left\{ \mathbb{1}(\tilde{\Phi}_{n,T} \leq x) - \mathbb{1}(\Phi_{n,T} \leq x) \right\} \mathbb{1}(|\tilde{\Phi}_{n,T} - \Phi_{n,T}| \leq \delta_T) \right] \right| \\
& \leq \sup_{x \in \mathbb{R}} \mathbb{E} \left[\mathbb{1}(|\Phi_{n,T} - x| \leq \delta_T, |\tilde{\Phi}_{n,T} - \Phi_{n,T}| \leq \delta_T) \right] \\
& \leq \sup_{x \in \mathbb{R}} \mathbb{P}(|\Phi_{n,T} - x| \leq \delta_T) = o(1)
\end{aligned}$$

by Step 2, we arrive at (C.5). $\square$

Step 4

In this step, we show that the auxiliary statistic $\widehat{\widehat{\Phi}}_{n,T}$ is close to $\widehat{\Phi}_{n,T}$ in the following sense.

Proposition C.9. *It holds that*

$$\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T} = o_p(\delta_T)$$

with $\delta_T = T^{1/q} / \sqrt{Th_{\min}} + \rho_T \sqrt{\log T}$.

The proof can be found at the end of this section.

Step 5

We finally show that

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(\widehat{\widehat{\Phi}}_{n,T} \leq x) - \mathbb{P}(\widehat{\Phi}_{n,T} \leq x)| = o(1). \quad (\text{C.6})$$

Proof of (C.6). To start with, we verify that for any $x \in \mathbb{R}$ and any $\delta > 0$,

$$\begin{aligned}
& \mathbb{P}(\widehat{\widehat{\Phi}}_{n,T} \leq x - \delta) - \mathbb{P}(|\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| > \delta) \\
& \leq \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) \leq \mathbb{P}(\widehat{\widehat{\Phi}}_{n,T} \leq x + \delta) + \mathbb{P}(|\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| > \delta).
\end{aligned} \quad (\text{C.7})$$

It holds that

$$\begin{aligned}
\mathbb{P}(\widehat{\Phi}_{n,T} \leq x) &= \mathbb{P}(\widehat{\Phi}_{n,T} \leq x, |\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| \leq \delta) + \mathbb{P}(\widehat{\Phi}_{n,T} \leq x, |\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| > \delta) \\
&\leq \mathbb{P}(\widehat{\Phi}_{n,T} \leq x, \widehat{\Phi}_{n,T} - \delta \leq \widehat{\widehat{\Phi}}_{n,T} \leq \widehat{\Phi}_{n,T} + \delta) + \mathbb{P}(|\widehat{\widehat{\Phi}}_{n,T} - \widehat{\Phi}_{n,T}| > \delta)
\end{aligned}$$

$$\leq \mathbb{P}(\widehat{\Phi}_{n,T} \leq x + \delta) + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta)$$

and analogously

$$\mathbb{P}(\widehat{\Phi}_{n,T} \leq x - \delta) \leq \mathbb{P}(\Phi_{n,T} \leq x) + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta).$$

Combining these two inequalities, we arrive at (C.7).

Now let $x \in \mathbb{R}$ be any point such that $\mathbb{P}(\widehat{\Phi}_{n,T} \leq x) \geq \mathbb{P}(\Phi_{n,T} \leq x)$. With the help of (C.7), we get that

$$\begin{aligned} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| &= \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \\ &\leq \mathbb{P}(\widehat{\Phi}_{n,T} \leq x + \delta_T) + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T) \\ &\quad - \mathbb{P}(\Phi_{n,T} \leq x) \\ &= \mathbb{P}(\widehat{\Phi}_{n,T} \leq x + \delta_T) - \mathbb{P}(\Phi_{n,T} \leq x + \delta_T) \\ &\quad + \mathbb{P}(\Phi_{n,T} \leq x + \delta_T) - \mathbb{P}(\Phi_{n,T} \leq x) \\ &\quad + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T). \end{aligned}$$

Analogously, for any point $x \in \mathbb{R}$ with $\mathbb{P}(\widehat{\Phi}_{n,T} \leq x) < \mathbb{P}(\Phi_{n,T} \leq x)$, it holds that

$$\begin{aligned} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| &\leq \mathbb{P}(\Phi_{n,T} \leq x - \delta_T) - \mathbb{P}(\widehat{\Phi}_{n,T} \leq x - \delta_T) \\ &\quad + \mathbb{P}(\Phi_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x - \delta_T) \\ &\quad + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T). \end{aligned}$$

Consequently,

$$\begin{aligned} \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| &\leq \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| \\ &\quad + \sup_{x \in \mathbb{R}} \mathbb{P}(|\Phi_{n,T} - x| \leq \delta_T) \\ &\quad + \mathbb{P}(|\widehat{\Phi}_{n,T} - \Phi_{n,T}| > \delta_T). \end{aligned}$$

Since the terms on the right-hand side are all $o(1)$ by Steps 2–4, we arrive at (C.6). $\square$

Details on Steps 1–5

Proof of Proposition C.7. Consider the stationary process $\mathcal{E}_i = \{\varepsilon_{it} : 1 \leq t \leq T\}$ for some fixed $i \in \{1, \dots, n\}$. By Theorem 2.1 and Corollary 2.1 in Berkes et al. (2014), the following strong approximation result holds true: On a richer probability space, there exist a standard Brownian motion $\mathbb{B}_i$ and a sequence $\{\tilde{\varepsilon}_{it} : t \in \mathbb{N}\}$ such that $[\tilde{\varepsilon}_{i1}, \dots, \tilde{\varepsilon}_{iT}] \stackrel{\mathcal{D}}{=} [\varepsilon_{i1}, \dots, \varepsilon_{iT}]$ for each T and

$$\max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \mathbb{B}_i(t) \right| = o(T^{1/q}) \quad \text{a.s.}, \quad (\text{C.8})$$

where $\sigma_i^2 = \sum_{k \in \mathbb{Z}} \text{Cov}(\varepsilon_{i0}, \varepsilon_{ik})$ denotes the long-run error variance. We apply this result separately for each $i \in \{1, \dots, n\}$. Since the error processes $\mathcal{E}_i = \{\varepsilon_{it} : 1 \leq t \leq T\}$ are independent across i , we can construct the processes $\tilde{\mathcal{E}}_i = \{\tilde{\varepsilon}_{it} : t \in \mathbb{N}\}$ in such a way that they are independent across i as well.

We now define the statistic $\tilde{\Phi}_{n,T}$ in the same way as $\hat{\Phi}_{n,T}$ except that the error processes $\mathcal{E}_i$ are replaced by $\tilde{\mathcal{E}}_i$. Specifically, we set

$$\tilde{\Phi}_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\tilde{\phi}_{ij,T}(u,h)}{(\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2)^{1/2}} \right| - \lambda(h) \right\},$$

where

$$\tilde{\phi}_{ij,T}(u,h) = \sum_{t=1}^T w_{t,T}(u,h) \{(\tilde{\varepsilon}_{it} - \tilde{\varepsilon}_i) - (\tilde{\varepsilon}_{jt} - \tilde{\varepsilon}_j)\}$$

and the estimator $\tilde{\sigma}_i^2$ is constructed from the sample $\tilde{\mathcal{E}}_i$ in the same way as $\hat{\sigma}_i^2$ is constructed from $\mathcal{E}_i$. Since $[\tilde{\varepsilon}_{i1}, \dots, \tilde{\varepsilon}_{iT}] \stackrel{\mathcal{D}}{=} [\varepsilon_{i1}, \dots, \varepsilon_{iT}]$ and $\hat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$, we have that $\tilde{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$ as well. In addition to $\tilde{\Phi}_{n,T}$, we introduce the Gaussian statistic

$$\Phi_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\phi_{ij,T}(u,h)}{(\sigma_i^2 + \sigma_j^2)^{1/2}} \right| - \lambda(h) \right\}$$

and the auxiliary statistic

$$\Phi_{n,T}^\diamond = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\phi_{ij,T}(u,h)}{(\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2)^{1/2}} \right| - \lambda(h) \right\},$$

where $\phi_{ij,T}(u,h) = \sum_{t=1}^T w_{t,T}(u,h) \{\sigma_i(Z_{it} - \bar{Z}_i) - \sigma_j(Z_{jt} - \bar{Z}_j)\}$ and the Gaussian variables

Z_{it} are chosen as $Z_{it} = \mathbb{B}_i(t) - \mathbb{B}_i(t-1)$. With this notation, we obtain the obvious bound

$$|\tilde{\Phi}_{n,T} - \Phi_{n,T}| \leq |\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond| + |\Phi_{n,T}^\diamond - \Phi_{n,T}|.$$

In what follows, we prove that

$$|\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond| = o_p\left(\frac{T^{1/q}}{\sqrt{Th_{\min}}}\right) \quad (\text{C.9})$$

$$|\Phi_{n,T}^\diamond - \Phi_{n,T}| = o_p(\rho_T \sqrt{\log T}), \quad (\text{C.10})$$

which completes the proof.

First consider $|\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond|$. Straightforward calculations yield that

$$\begin{aligned} |\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond| &\leq \max_{1 \leq i < j \leq n} (\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2)^{-1/2} \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)| \\ &= O_p(1) \cdot \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)|, \end{aligned} \quad (\text{C.11})$$

where the last line follows from the fact that $\tilde{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$. Using summation by parts (that is, $\sum_{t=1}^T a_t b_t = \sum_{t=1}^{T-1} A_t(b_t - b_{t+1}) + A_T b_T$ with $A_t = \sum_{s=1}^t a_s$), we further obtain that

$$\begin{aligned} &|\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)| \\ &= \left| \sum_{t=1}^T w_{t,T}(u,h) \{(\tilde{\varepsilon}_{it} - \tilde{\varepsilon}_i) - (\tilde{\varepsilon}_{jt} - \tilde{\varepsilon}_j) - \sigma_i(Z_{it} - \bar{Z}_i) + \sigma_j(Z_{jt} - \bar{Z}_j)\} \right| \\ &= \left| \sum_{t=1}^{T-1} A_{ij,t}(w_{t,T}(u,h) - w_{t+1,T}(u,h)) + A_{ij,T} w_{T,T}(u,h) \right|, \end{aligned}$$

where

$$A_{ij,t} = \sum_{s=1}^t \{(\tilde{\varepsilon}_{is} - \tilde{\varepsilon}_i) - (\tilde{\varepsilon}_{js} - \tilde{\varepsilon}_j) - \sigma_i(Z_{is} - \bar{Z}_i) + \sigma_j(Z_{js} - \bar{Z}_j)\}$$

and $A_{ij,T} = 0$ for all pairs (i,j) by construction. From this, it follows that

$$|\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)| \leq W_T(u,h) \max_{1 \leq t \leq T} |A_{ij,t}| \quad (\text{C.12})$$

with $W_T(u,h) = \sum_{t=1}^{T-1} |w_{t+1,T}(u,h) - w_{t,T}(u,h)|$. Straightforward calculations yield that

$$\begin{aligned} \max_{1 \leq t \leq T} |A_{ij,t}| &\leq \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \sum_{s=1}^t Z_{is} \right| + \max_{1 \leq t \leq T} |t(\tilde{\varepsilon}_i - \sigma_i \bar{Z}_i)| \\ &\quad + \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{js} - \sigma_j \sum_{s=1}^t Z_{js} \right| + \max_{1 \leq t \leq T} |t(\tilde{\varepsilon}_j - \sigma_j \bar{Z}_j)| \end{aligned}$$

$$\begin{aligned}
&\leq 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \sum_{s=1}^t Z_{is} \right| + 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{js} - \sigma_j \sum_{s=1}^t Z_{js} \right| \\
&= 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \sum_{s=1}^t (\mathbb{B}_i(s) - \mathbb{B}_i(s-1)) \right| \\
&\quad + 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{js} - \sigma_j \sum_{s=1}^t (\mathbb{B}_j(s) - \mathbb{B}_j(s-1)) \right| \\
&= 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{is} - \sigma_i \mathbb{B}_i(t) \right| + 2 \max_{1 \leq t \leq T} \left| \sum_{s=1}^t \tilde{\varepsilon}_{js} - \sigma_j \mathbb{B}_j(t) \right|.
\end{aligned}$$

Applying the strong approximation result (C.8), we can infer that

$$\max_{1 \leq t \leq T} |A_{ij,t}| = o_p(T^{1/q}).$$

Moreover, standard arguments show that $\max_{(u,h) \in \mathcal{G}_T} W_T(u,h) = O(1/\sqrt{Th_{\min}})$. Plugging these two results into (C.12), we obtain that

$$\max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u,h) - \phi_{ij,T}(u,h)| = o_p\left(\frac{T^{1/q}}{\sqrt{Th_{\min}}}\right),$$

which in view of (C.11) yields that $|\tilde{\Phi}_{n,T} - \Phi_{n,T}^\diamond| = o_p(T^{1/q}/\sqrt{Th_{\min}})$. This completes the proof of (C.9).

Next consider $|\Phi_{n,T}^\diamond - \Phi_{n,T}|$. It holds that

$$\begin{aligned}
|\Phi_{n,T}^\diamond - \Phi_{n,T}| &\leq \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\phi_{ij,T}(u,h)}{\{\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2\}^{1/2}} - \frac{\phi_{ij,T}(u,h)}{\{\sigma_i^2 + \sigma_j^2\}^{1/2}} \right| \\
&\leq \max_{1 \leq i < j \leq n} \left\{ \left| (\tilde{\sigma}_i^2 + \tilde{\sigma}_j^2)^{-1/2} - (\sigma_i^2 + \sigma_j^2)^{-1/2} \right| \right\} \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\phi_{ij,T}(u,h)| \\
&= o_p(\rho_T) \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} |\phi_{ij,T}(u,h)|, \tag{C.13}
\end{aligned}$$

where the last line is due to the fact that $\tilde{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$. We can write $\phi_{ij,T}(u,h) = \phi_{ij,T}^{(I)}(u,h) - \phi_{ij,T}^{(II)}(u,h)$, where

$$\begin{aligned}
\phi_{ij,T}^{(I)}(u,h) &= \sum_{t=1}^T w_{t,T}(u,h) (\sigma_i Z_{it} - \sigma_j Z_{jt}) \sim N(0, \sigma_i^2 + \sigma_j^2) \\
\phi_{ij,T}^{(II)}(u,h) &= \sum_{t=1}^T w_{t,T}(u,h) (\sigma_i \bar{Z}_i - \sigma_j \bar{Z}_j) \sim N(0, (\sigma_i^2 + \sigma_j^2) c_T(u,h))
\end{aligned}$$

with $c_T(u,h) = \{\sum_{t=1}^T w_{t,T}(u,h)\}^2/T \leq C < \infty$ for all $(u,h) \in \mathcal{G}_T$ and $1 \leq i < j \leq n$. This shows that $\phi_{ij,T}(u,h)$ are centred Gaussian random variables with bounded variance for all

$(u, h) \in \mathcal{G}_T$ and $1 \leq i < j \leq n$. Hence, standard results on the maximum of Gaussian random variables yield that

$$\max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} |\phi_{ij,T}(u, h)| = O_p(\sqrt{\log T}), \quad (\text{C.14})$$

where we have used that n is fixed and $|\mathcal{G}_T| = O(T^\theta)$ for some large but fixed constant θ by Assumption (C10). Plugging this into (C.13) yields $|\Phi_{n,T}^\diamond - \Phi_{n,T}| = o_p(\rho_T \sqrt{\log T})$, which completes the proof of (C.10). $\square$

Proof of Proposition C.8. The proof is an application of anti-concentration bounds for Gaussian random vectors. We in particular make use of the following anti-concentration inequality from Nazarov (2003), which can also be found as Lemma A.1 in Chernozhukov et al. (2017).

Lemma C.10. *Let $\mathbf{Z} = (Z_1, \dots, Z_p)^\top$ be a centred Gaussian random vector in $\mathbb{R}^p$ such that $\mathbb{E}[Z_j^2] \geq b$ for all $1 \leq j \leq p$ and some constant $b > 0$. Then for every $\mathbf{z} \in \mathbb{R}^p$ and $a > 0$,*

$$\mathbb{P}(\mathbf{Z} \leq \mathbf{z} + a) - \mathbb{P}(\mathbf{Z} \leq \mathbf{z}) \leq Ca\sqrt{\log p},$$

where the constant C only depends on b .

To apply this result, we introduce the following notation: We write $x = (u, h)$ and $\mathcal{G}_T = \{x_1, \dots, x_p\}$, where $p := |\mathcal{G}_T| \leq O(T^\theta)$ for some large but fixed $\theta > 0$ by our assumptions.

For $k = 1, \dots, p$ and $1 \leq i < j \leq n$, we further let

$$Z_{ij,2k-1} = \frac{\phi_{ij,T}(x_{k1}, x_{k2})}{\{\sigma_i^2 + \sigma_j^2\}^{1/2}} \quad \text{and} \quad Z_{ij,2k} = -\frac{\phi_{ij,T}(x_{k1}, x_{k2})}{\{\sigma_i^2 + \sigma_j^2\}^{1/2}}$$

along with $\lambda_{ij,2k-1} = \lambda(x_{k2})$ and $\lambda_{ij,2k} = \lambda(x_{k2})$, where $x_k = (x_{k1}, x_{k2})$. Under our assumptions, it holds that $\mathbb{E}[Z_{ij,l}] = 0$ and $\mathbb{E}[Z_{ij,l}^2] \geq b > 0$ for all i, j and l . We next construct the random vector $\mathbf{Z} = (Z_{ij,l} : 1 \leq i < j \leq n, 1 \leq l \leq 2p)$ by stacking the variables $Z_{ij,l}$ in a certain order (which can be chosen freely) and construct the vector $\boldsymbol{\lambda} = (\lambda_{ij,l} : 1 \leq i < j \leq n, 1 \leq l \leq 2p)$ in an analogous way. Since the variables $Z_{ij,l}$ are

normally distributed, $\mathbf{Z}$ is a Gaussian random vector of length $(n-1)np$.

With this notation at hand, we can express the probability $\mathbb{P}(\Phi_{n,T} \leq q)$ as follows for each $q \in \mathbb{R}$:

$$\begin{aligned}\mathbb{P}(\Phi_{n,T} \leq q) &= \mathbb{P}\left(\max_{1 \leq i < j \leq n} \max_{1 \leq l \leq 2p} \{Z_{ij,l} - \lambda_{ij,l}\} \leq q\right) \\ &= \mathbb{P}(Z_{ij,l} \leq \lambda_{ij,l} + q \text{ for all } (i, j, l)) = \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + q).\end{aligned}$$

Consequently,

$$\begin{aligned}\mathbb{P}(|\Phi_{n,T} - x| \leq \delta_T) &= \mathbb{P}(x - \delta_T \leq \Phi_{n,T} \leq x + \delta_T) \\ &= \mathbb{P}(\Phi_{n,T} \leq x + \delta_T) - \mathbb{P}(\Phi_{n,T} \leq x) \\ &\quad + \mathbb{P}(\Phi_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x - \delta_T) \\ &= \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + x + \delta_T) - \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + x) \\ &\quad + \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + x) - \mathbb{P}(\mathbf{Z} \leq \boldsymbol{\lambda} + x - \delta_T) \\ &\leq 2C\delta_T \sqrt{\log((n-1)np)},\end{aligned}$$

where the last line is by Lemma C.10. This immediately implies Proposition C.8. $\square$

Proof of Proposition C.9. Straightforward calculations yield that

$$\begin{aligned}|\widehat{\Phi}_{n,T} - \widehat{\Phi}_{n,T}| &\leq \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} - \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| \\ &\quad + \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} - \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right|.\end{aligned}$$

Since $\widehat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$ and $\widehat{\sigma}_j^2 = \sigma_j^2 + o_p(\rho_T)$, we further get that

$$\begin{aligned}&\max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} - \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| \\ &\leq \max_{1 \leq i < j \leq n} \left\{ \left| (\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{-1/2} - (\sigma_i^2 + \sigma_j^2)^{-1/2} \right| \right\} \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) \right| \\ &= o_p(\rho_T) \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) \right|\end{aligned}$$

and

$$\begin{aligned}
& \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} - \frac{\widehat{\phi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| \\
& \leq \max_{1 \leq i < j \leq n} \left\{ (\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{-1/2} \right\} \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) - \widehat{\phi}_{ij,T}(u,h) \right| \\
& = O_p(1) \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) - \widehat{\phi}_{ij,T}(u,h) \right|,
\end{aligned}$$

where the difference of the kernel averages $\widehat{\phi}_{ij,T}(u,h) - \widehat{\phi}_{ij,T}(u,h)$ does not include the error terms (they cancel out) and can be written as

$$\begin{aligned}
& \left| \widehat{\phi}_{ij,T}(u,h) - \widehat{\phi}_{ij,T}(u,h) \right| \\
& = \left| \sum_{t=1}^T w_{t,T}(u,h) \{ (\beta_i - \widehat{\beta}_i)^\top (\mathbf{X}_{it} - \bar{\mathbf{X}}_i) - (\beta_j - \widehat{\beta}_j)^\top (\mathbf{X}_{jt} - \bar{\mathbf{X}}_j) \} \right| \\
& \leq \left| (\beta_i - \widehat{\beta}_i)^\top \sum_{t=1}^T w_{t,T}(u,h) \mathbf{X}_{it} \right| + \left| (\beta_i - \widehat{\beta}_i)^\top \bar{\mathbf{X}}_i \right| \left| \sum_{t=1}^T w_{t,T}(u,h) \right| \\
& \quad + \left| (\beta_j - \widehat{\beta}_j)^\top \sum_{t=1}^T w_{t,T}(u,h) \mathbf{X}_{jt} \right| + \left| (\beta_j - \widehat{\beta}_j)^\top \bar{\mathbf{X}}_j \right| \left| \sum_{t=1}^T w_{t,T}(u,h) \right|.
\end{aligned}$$

Hence,

$$\left| \widehat{\Phi}_{n,T} - \widehat{\Phi}_{n,T} \right| \leq o_p(\rho_T) A_{n,T} + O_p(1) \{ 2B_{n,T} + 2C_{n,T} \}, \quad (\text{C.15})$$

where

$$\begin{aligned}
A_{n,T} &= \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) \right| \\
B_{n,T} &= \max_{1 \leq i \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| (\beta_i - \widehat{\beta}_i)^\top \sum_{t=1}^T w_{t,T}(u,h) \mathbf{X}_{it} \right| \\
C_{n,T} &= \max_{1 \leq i \leq n} \left| (\beta_i - \widehat{\beta}_i)^\top \bar{\mathbf{X}}_i \right| \max_{(u,h) \in \mathcal{G}_T} \left| \sum_{t=1}^T w_{t,T}(u,h) \right|.
\end{aligned}$$

We examine these three terms separately.

We first prove that

$$A_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| \widehat{\phi}_{ij,T}(u,h) \right| = O_p(\sqrt{\log T}). \quad (\text{C.16})$$

From the proof of Proposition C.7, we know that there exist identically distributed versions

$\tilde{\phi}_{ij,T}(u, h)$ of the statistics $\widehat{\phi}_{ij,T}(u, h)$ with the property that

$$\max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u, h) - \phi_{ij,T}(u, h)| = o_p\left(\frac{T^{1/q}}{\sqrt{Th_{\min}}}\right). \quad (\text{C.17})$$

Instead of (C.16), it thus suffices to show that

$$\max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} |\tilde{\phi}_{ij,T}(u, h)| = O_p(\sqrt{\log T}). \quad (\text{C.18})$$

Since for any constant $c > 0$,

$$\begin{aligned} & \mathbb{P}\left(\max_{i,j,(u,h)} |\phi_{ij,T}(u, h)| \leq \frac{c\sqrt{\log T}}{2}\right) \\ & \leq \mathbb{P}\left(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| \leq c\sqrt{\log T}\right) \\ & \quad + \mathbb{P}\left(\left|\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| - \max_{i,j,(u,h)} |\phi_{ij,T}(u, h)|\right| > \frac{c\sqrt{\log T}}{2}\right) \\ & \leq \mathbb{P}\left(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| \leq c\sqrt{\log T}\right) \\ & \quad + \mathbb{P}\left(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h) - \phi_{ij,T}(u, h)| > \frac{c\sqrt{\log T}}{2}\right) \end{aligned}$$

and $\mathbb{P}(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h) - \phi_{ij,T}(u, h)| > c\sqrt{\log T}/2) = o(1)$ by (C.17), we get that

$$\begin{aligned} & \mathbb{P}\left(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| \leq c\sqrt{\log T}\right) \\ & \geq \mathbb{P}\left(\max_{i,j,(u,h)} |\phi_{ij,T}(u, h)| \leq \frac{c\sqrt{\log T}}{2}\right) - o(1). \end{aligned} \quad (\text{C.19})$$

Moreover, since $\max_{i,j,(u,h)} |\phi_{ij,T}(u, h)| = O_p(\sqrt{\log T})$ as already proven in (C.14), we can make the probability $\mathbb{P}(\max_{i,j,(u,h)} |\phi_{ij,T}(u, h)| \leq c\sqrt{\log T}/2)$ on the right-hand side of (C.19) arbitrarily close to 1 by choosing the constant c sufficiently large. Hence, for any $\delta > 0$, we can find a constant $c > 0$ such that $\mathbb{P}(\max_{i,j,(u,h)} |\tilde{\phi}_{ij,T}(u, h)| \leq c\sqrt{\log T}) \geq 1 - \delta$ for sufficiently large T . This proves (C.18), which in turn yields (C.16).

We next turn to $B_{n,T}$. Without loss of generality, we assume that $\mathbf{X}_{it}$ is real-valued. The vector-valued case can be handled analogously. To start with, we have a closer look at the term $\sum_{t=1}^T w_{t,T}(u, h) \mathbf{X}_{it}$. By construction, the kernel weights $w_{t,T}(u, h)$ are unequal to 0 if

and only if $T(u - h) \leq t \leq T(u + h)$. We can use this fact to write

$$\left| \sum_{t=1}^T w_{t,T}(u, h) \mathbf{X}_{it} \right| = \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}(u, h) \mathbf{X}_{it} \right|.$$

Note that

$$\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}^2(u, h) = \sum_{t=1}^T w_{t,T}^2(u, h) = \sum_{t=1}^T \frac{\Lambda_{t,T}^2(u, h)}{\sum_{s=1}^T \Lambda_{s,T}^2(u, h)} = 1. \quad (\text{C.20})$$

Denoting by $D_{T,u,h}$ the number of integers between $\lfloor T(u - h) \rfloor$ and $\lceil T(u + h) \rceil$ (with the obvious bounds $2Th \leq D_{T,u,h} \leq 2Th + 2$) and using (C.20), we can normalize the kernel weights as follows:

$$\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} (\sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h))^2 = D_{T,u,h}.$$

Next, we apply Proposition C.2 with the weights $a_t = \sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h)$ to obtain that

$$\begin{aligned} & \mathbb{P} \left(\left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} \sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h) \mathbf{X}_{it} \right| \geq x \right) \\ & \leq C_1 \frac{(\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} |\sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h)|^{q'}) \|\mathbf{X}_i\|_{q',\alpha}^{q'}}{x^{q'}} \\ & \quad + C_2 \exp \left(-\frac{C_3 x^2}{D_{T,u,h} \|\mathbf{X}_i\|_{2,\alpha}^2} \right) \end{aligned} \quad (\text{C.21})$$

for any $x > 0$, where $\|\mathbf{X}_i\|_{q',\alpha}^{q'} = \sup_{t \geq 0} (t+1)^\alpha \sum_{s=t}^\infty \delta_{q'}(\mathbf{h}_i, s)$ is the dependence adjusted norm introduced in Definition C.1 and $\|\mathbf{X}_i\|_{q',\alpha}^{q'} < \infty$ by Assumption (C6). From (C.21), it follows that for any $\delta > 0$,

$$\begin{aligned} & \mathbb{P} \left(\max_{(u,h) \in \mathcal{G}_T} \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}(u, h) \mathbf{X}_{it} \right| \geq \delta T^{1/q} \right) \\ & \leq \sum_{(u,h) \in \mathcal{G}_T} \mathbb{P} \left(\left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}(u, h) \mathbf{X}_{it} \right| \geq \delta T^{1/q} \right) \\ & = \sum_{(u,h) \in \mathcal{G}_T} \mathbb{P} \left(\left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} \sqrt{D_{T,u,h}} \cdot w_{t,T}(u, h) \mathbf{X}_{it} \right| \geq \delta \sqrt{D_{T,u,h}} T^{1/q} \right) \\ & \leq \sum_{(u,h) \in \mathcal{G}_T} \left[C_1 \frac{(\sqrt{D_{T,u,h}})^{q'} (\sum |w_{t,T}(u, h)|^{q'}) \|\mathbf{X}_i\|_{q',\alpha}^{q'}}{(\delta \sqrt{D_{T,u,h}} T^{1/q})^{q'}} + C_2 \exp \left(-\frac{C_3 (\delta \sqrt{D_{T,u,h}} T^{1/q})^2}{D_{T,u,h} \|\mathbf{X}_i\|_{2,\alpha}^2} \right) \right] \end{aligned}$$

$$\begin{aligned}
&= \sum_{(u,h) \in \mathcal{G}_T} \left[C_1 \frac{(\sum |w_{t,T}(u,h)|^{q'}) \|\mathbf{X}_{i\cdot}\|_{q',\alpha}^{q'}}{\delta^{q'} T^{q'/q}} + C_2 \exp\left(-\frac{C_3 \delta^2 T^{2/q}}{\|\mathbf{X}_{i\cdot}\|_{2,\alpha}^2}\right) \right] \\
&\leq C_1 \frac{T^\theta \|\mathbf{X}_{i\cdot}\|_{q',\alpha}^{q'}}{\delta^{q'} T^{q'/q}} \max_{(u,h) \in \mathcal{G}_T} \left(\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} |w_{t,T}(u,h)|^{q'} \right) + C_2 T^\theta \exp\left(-\frac{C_3 \delta^2 T^{2/q}}{\|\mathbf{X}_{i\cdot}\|_{2,\alpha}^2}\right) \\
&= C \frac{T^{\theta-q'/q}}{\delta^{q'}} + C T^\theta \exp(-C T^{2/q} \delta^2), \tag{C.22}
\end{aligned}$$

where the constant C depends neither on T nor on δ . In the last equality of the above display, we have used the following facts:

- (i) $\|\mathbf{X}_{i\cdot}\|_{q',\alpha}^{q'} < \infty$ by Assumption (C6).
- (ii) $\|\mathbf{X}_{i\cdot}\|_{2,\alpha}^2 < \infty$ (which follows from (i)).
- (iii) $\max_{(u,h) \in \mathcal{G}_T} (\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} |w_{t,T}(u,h)|^{q'}) \leq 1$ for the following reason: By (C.20), it holds that $\sum_{t=1}^T w_{t,T}^2(u,h) = 1$ and thus $0 \leq w_{t,T}^2(u,h) \leq 1$ for all t, T and (u,h) . This implies that $0 \leq |w_{t,T}(u,h)|^{q'} = (w_{t,T}^2(u,h))^{q'/2} \leq w_{t,T}^2(u,h) \leq 1$ for all t, T and (u,h) .

As a result,

$$\max_{(u,h) \in \mathcal{G}_T} \left(\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} |w_{t,T}(u,h)|^{q'} \right) \leq \max_{(u,h) \in \mathcal{G}_T} \left(\sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}^2(u,h) \right) = 1.$$

Since $\theta - q'/q < 0$ by Assumption (C4), the bound in (C.22) converges to 0 as $T \rightarrow \infty$ for any fixed $\delta > 0$. Consequently, we obtain that

$$\max_{(u,h) \in \mathcal{G}_T} \left| \sum_{t=\lfloor T(u-h) \rfloor}^{\lceil T(u+h) \rceil} w_{t,T}(u,h) \mathbf{X}_{it} \right| = o_p(T^{1/q}). \tag{C.23}$$

Using this together with the fact that $\beta_i - \hat{\beta}_i = O_p(1/\sqrt{T})$ (which is the statement of Lemma C.5), we arrive at the bound

$$B_{n,T} = \max_{1 \leq i \leq n} \max_{(u,h) \in \mathcal{G}_T} \left| (\beta_i - \hat{\beta}_i)^\top \sum_{t=1}^T w_{t,T}(u,h) \mathbf{X}_{it} \right| = o_p\left(\frac{T^{1/q}}{\sqrt{T}}\right).$$

We finally turn to $C_{n,T}$. By straightforward calculations, $|\sum_{t=1}^T w_{t,T}(u,h)| \leq C\sqrt{Th_{\max}} = o(\sqrt{T})$. Moreover, $\bar{\mathbf{X}}_i = O_p(1/\sqrt{T})$ by Lemma C.3 and $\beta_i - \hat{\beta}_i = O_p(1/\sqrt{T})$ by Lemma

C.5. This immediately yields that

$$C_{n,T} = \max_{1 \leq i \leq n} |(\beta_i - \hat{\beta}_i)^\top \bar{\mathbf{X}}_i| \max_{(u,h) \in \mathcal{G}_T} \left| \sum_{t=1}^T w_{t,T}(u, h) \right| = o_p\left(\frac{1}{\sqrt{T}}\right).$$

To summarize, we have shown that

$$\begin{aligned} |\hat{\Phi}_{n,T} - \hat{\Phi}_{n,T}| &\leq o_p(\rho_T) A_{n,T} + O_p(1) \{2B_{n,T} + 2C_{n,T}\} \\ &= o_p(\rho_T) O_p(\sqrt{\log T}) + o_p\left(\frac{T^{1/q}}{\sqrt{T}}\right) + o_p\left(\frac{1}{\sqrt{T}}\right). \end{aligned}$$

This immediately implies the desired result. $\square$

Proof of Proposition 4.4

We first show that

$$\mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) = 1 - \alpha. \quad (\text{C.24})$$

We proceed by contradiction. Suppose that (C.24) does not hold true. Since $\mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) \geq 1 - \alpha$ by definition of the quantile $q_{n,T}(\alpha)$, there exists $\xi > 0$ such that $\mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) = 1 - \alpha + \xi$. From the proof of Proposition C.8, we know that for any $\delta > 0$,

$$\begin{aligned} &\mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) - \mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha) - \delta) \\ &\leq \sup_{x \in \mathbb{R}} \mathbb{P}(|\Phi_{n,T} - x| \leq \delta) \leq 2C\delta\sqrt{\log((n-1)np)}. \end{aligned}$$

Hence,

$$\begin{aligned} \mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha) - \delta) &\geq \mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) - 2C\delta\sqrt{\log((n-1)np)} \\ &= 1 - \alpha + \xi - 2C\delta\sqrt{\log((n-1)np)} > 1 - \alpha \end{aligned}$$

for $\delta > 0$ small enough. This contradicts the definition of the quantile $q_{n,T}(\alpha)$ according to which $q_{n,T}(\alpha) = \inf_{q \in \mathbb{R}} \{\mathbb{P}(\Phi_{n,T} \leq q) \geq 1 - \alpha\}$. We thus arrive at (C.24).

Proposition 4.4 is a simple consequence of Theorem 4.1 and equation (C.24). Specifically, we obtain that under H_0 ,

$$|\mathbb{P}(\hat{\Psi}_{n,T} \leq q_{n,T}(\alpha)) - (1 - \alpha)| = |\mathbb{P}(\hat{\Phi}_{n,T} \leq q_{n,T}(\alpha)) - (1 - \alpha)|$$

$$\begin{aligned}
&= \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq q_{n,T}(\alpha)) - \mathbb{P}(\Phi_{n,T} \leq q_{n,T}(\alpha)) \right| \\
&\leq \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\widehat{\Phi}_{n,T} \leq x) - \mathbb{P}(\Phi_{n,T} \leq x) \right| = o(1).
\end{aligned}$$

Proof of Proposition 4.5

To start with, note that for some sufficiently large constant C we have

$$\lambda(h) = \sqrt{2 \log\{1/(2h)\}} \leq \sqrt{2 \log\{1/(2h_{\min})\}} \leq C \sqrt{\log T}. \quad (\text{C.25})$$

Write $\widehat{\psi}_{ij,T}(u, h) = \widehat{\psi}_{ij,T}^A(u, h) + \widehat{\psi}_{ij,T}^B(u, h)$ with

$$\begin{aligned}
\widehat{\psi}_{ij,T}^A(u, h) &= \sum_{t=1}^T w_{t,T}(u, h) \left\{ (\varepsilon_{it} - \bar{\varepsilon}_i) + (\beta_i - \widehat{\beta}_i)^\top (\mathbf{X}_{it} - \bar{\mathbf{X}}_i) - \bar{m}_{i,T} \right. \\
&\quad \left. - (\varepsilon_{jt} - \bar{\varepsilon}_j) - (\beta_j - \widehat{\beta}_j)^\top (\mathbf{X}_{jt} - \bar{\mathbf{X}}_j) + \bar{m}_{j,T} \right\} \\
\widehat{\psi}_{ij,T}^B(u, h) &= \sum_{t=1}^T w_{t,T}(u, h) \left(m_{i,T}\left(\frac{t}{T}\right) - m_{j,T}\left(\frac{t}{T}\right) \right),
\end{aligned}$$

where $\bar{m}_{i,T} = T^{-1} \sum_{t=1}^T m_{i,T}(t/T)$. Without loss of generality, consider the following scenario: there exists $(u_0, h_0) \in \mathcal{G}_T$ with $[u_0 - h_0, u_0 + h_0] \subseteq [0, 1]$ such that

$$m_{i,T}(w) - m_{j,T}(w) \geq c_T \sqrt{\log T / (Th_0)} \quad (\text{C.26})$$

for all $w \in [u_0 - h_0, u_0 + h_0]$.

We first derive a lower bound on the term $\widehat{\psi}_{ij,T}^B(u_0, h_0)$. Since the kernel K is symmetric and $u_0 = t/T$ for some t , it holds that $S_{T,1}(u_0, h_0) = 0$ and thus,

$$\begin{aligned}
w_{t,T}(u_0, h_0) &= \frac{K\left(\frac{\frac{t}{T} - u_0}{h_0}\right) S_{T,2}(u_0, h_0)}{\left\{ \sum_{t=1}^T K^2\left(\frac{\frac{t}{T} - u_0}{h_0}\right) S_{T,2}^2(u_0, h_0) \right\}^{1/2}} \\
&= \frac{K\left(\frac{\frac{t}{T} - u_0}{h_0}\right)}{\left\{ \sum_{t=1}^T K^2\left(\frac{\frac{t}{T} - u_0}{h_0}\right) \right\}^{1/2}} \geq 0.
\end{aligned}$$

Together with (C.26), this implies that

$$\widehat{\psi}_{ij,T}^B(u_0, h_0) \geq c_T \sqrt{\frac{\log T}{Th_0}} \sum_{t=1}^T w_{t,T}(u_0, h_0). \quad (\text{C.27})$$

Using the Lipschitz continuity of the kernel K , we can show by straightforward calculations

that for any $(u, h) \in \mathcal{G}_T$ and any natural number ℓ ,

$$\left| \frac{1}{Th} \sum_{t=1}^T K\left(\frac{\frac{t}{T} - u}{h}\right) \left(\frac{\frac{t}{T} - u}{h}\right)^\ell - \int_0^1 \frac{1}{h} K\left(\frac{w - u}{h}\right) \left(\frac{w - u}{h}\right)^\ell dw \right| \leq \frac{C}{Th}, \quad (\text{C.28})$$

where the constant C does not depend on u , h and T . With the help of (C.28), we obtain that for any $(u, h) \in \mathcal{G}_T$ with $[u - h, u + h] \subseteq [0, 1]$,

$$\left| \sum_{t=1}^T w_{t,T}(u, h) - \frac{\sqrt{Th}}{\kappa} \right| \leq \frac{C}{\sqrt{Th}}, \quad (\text{C.29})$$

where $\kappa = (\int K^2(\varphi) d\varphi)^{1/2}$ and the constant C does once again not depend on u , h and T . From (C.29), it follows that $\sum_{t=1}^T w_{t,T}(u, h) \geq \sqrt{Th}/(2\kappa)$ for sufficiently large T and any $(u, h) \in \mathcal{G}_T$ with $[u - h, u + h] \subseteq [0, 1]$. This together with (C.27) allows us to infer that

$$\hat{\psi}_{ij,T}^B(u_0, h_0) \geq \frac{c_T \sqrt{\log T}}{2\kappa} \quad (\text{C.30})$$

for sufficiently large T .

We next analyze $\hat{\psi}_{ij,T}^A(u_0, h_0)$, which can be expressed as $\hat{\psi}_{ij,T}^A(u_0, h_0) = \hat{\phi}_{ij,T}(u, h) + (\bar{m}_{j,T} - \bar{m}_{i,T}) \sum_{t=1}^T w_{t,T}(u, h)$. The proof of Proposition C.9 shows that

$$\max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \left| \hat{\phi}_{ij,T}(u, h) \right| = O_p(\sqrt{\log T}).$$

Using this together with the bounds $\bar{m}_{i,T} \leq C/T$ and $\sum_{t=1}^T w_{t,T}(u, h) \leq C\sqrt{T}$, we can infer that

$$\begin{aligned} & \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \left| \hat{\psi}_{ij,T}^A(u, h) \right| \\ &= \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \left| \hat{\phi}_{ij,T}(u, h) + (\bar{m}_{j,T} - \bar{m}_{i,T}) \sum_{t=1}^T w_{t,T}(u, h) \right| = O_p(\sqrt{\log T}). \end{aligned} \quad (\text{C.31})$$

With the help of (C.30), (C.31), (C.25) and the assumption that $\hat{\sigma}_i^2 = \sigma_i^2 + o_p(\rho_T)$, we finally arrive at

$$\begin{aligned} \hat{\Psi}_{n,T} &\geq \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \frac{|\hat{\psi}_{ij,T}^B(u, h)|}{\{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{1/2}} - \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \left\{ \frac{|\hat{\psi}_{ij,T}^A(u, h)|}{\{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{1/2}} + \lambda(h) \right\} \\ &= \max_{1 \leq i < j \leq n} \max_{(u, h) \in \mathcal{G}_T} \frac{|\hat{\psi}_{ij,T}^B(u, h)|}{\{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{1/2}} + O_p(\sqrt{\log T}) \\ &\geq \frac{c_T \sqrt{\log T}}{2\kappa} \min_{1 \leq i < j \leq n} \{\hat{\sigma}_i^2 + \hat{\sigma}_j^2\}^{-1/2} + O_p(\sqrt{\log T}) \end{aligned}$$

$$= \frac{c_T \sqrt{\log T}}{2\kappa} O_p(1) + O_p(\sqrt{\log T}). \quad (\text{C.32})$$

Since $q_{n,T}(\alpha) = O(\sqrt{\log T})$ for any fixed $\alpha \in (0, 1)$ and $c_T \rightarrow \infty$, (C.32) immediately implies that $\mathbb{P}(\widehat{\Psi}_{n,T} \leq q_{n,T}(\alpha)) = o(1)$.

Proof of Proposition 4.6

Denote by $\mathcal{M}_0$ the set of quadruples $(i, j, u, h) \in \{1 \dots, n\}^2 \times \mathcal{G}_T$ for which $H_0^{[i,j]}(u, h)$ is true. Then we can write the FWER as

$$\begin{aligned} \text{FWER}(\alpha) &= \mathbb{P}\left(\exists (i, j, u, h) \in \mathcal{M}_0 : \widehat{\psi}_{ij,T}^0(u, h) > q_{n,T}(\alpha)\right) \\ &= \mathbb{P}\left(\max_{(i,j,u,h) \in \mathcal{M}_0} \widehat{\psi}_{ij,T}^0(u, h) > q_{n,T}(\alpha)\right) \\ &= \mathbb{P}\left(\max_{(i,j,u,h) \in \mathcal{M}_0} \widehat{\phi}_{ij,T}^0(u, h) > q_{n,T}(\alpha)\right) \\ &\leq \mathbb{P}\left(\max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \widehat{\phi}_{ij,T}^0(u, h) > q_{n,T}(\alpha)\right) \\ &= \mathbb{P}(\widehat{\Phi}_{n,T} > q_{n,T}(\alpha)) = \alpha + o(1), \end{aligned}$$

where the third equality uses that $\widehat{\psi}_{ijk,T}^0 = \widehat{\phi}_{ijk,T}^0$ under $H_0^{[i,j]}(u, h)$.

Proof of Proposition 5.1

For the sake of brevity, we introduce the following notation. For each i and j , we define the statistic $\widehat{\Psi}_{ij,T} := \max_{(u,h) \in \mathcal{G}_T} \widehat{\psi}_{ij,T}^0(u, h)$ which can be interpreted as a distance measure between the two curves m_i and m_j on the whole interval $[0, 1]$. Using this notation, we can rewrite the dissimilarity measure defined in (5.1) as

$$\widehat{\Delta}(S, S') = \max_{\substack{i \in S, \\ j \in S'}} \widehat{\Psi}_{ij,T}.$$

Now consider the event

$$B_{n,T} = \left\{ \max_{1 \leq \ell \leq N} \max_{i,j \in G_\ell} \widehat{\Psi}_{ij,T} \leq q_{n,T}(\alpha) \text{ and } \min_{1 \leq \ell < \ell' \leq N} \min_{\substack{i \in G_\ell, \\ j \in G_{\ell'}}} \widehat{\Psi}_{ij,T} > q_{n,T}(\alpha) \right\}.$$

The term $\max_{1 \leq \ell \leq N} \max_{i,j \in G_\ell} \widehat{\Psi}_{ij,T}$ is the largest multiscale distance between two time series i and j from the same group, whereas $\min_{1 \leq \ell < \ell' \leq N} \min_{i \in G_\ell, j \in G_{\ell'}} \widehat{\Psi}_{ij,T}$ is the smallest

multiscale distance between two time series from two different groups. On the event $B_{n,T}$, it obviously holds that

$$\max_{1 \leq \ell \leq N} \max_{i,j \in G_\ell} \widehat{\Psi}_{ij,T} < \min_{1 \leq \ell < \ell' \leq N} \min_{\substack{i \in G_\ell, \\ j \in G_{\ell'}}} \widehat{\Psi}_{ij,T}. \quad (\text{C.33})$$

Hence, any two time series from the same class have a smaller distance than any two time series from two different classes. With the help of Proposition 4.4, it is easy to see that

$$\mathbb{P}\left(\max_{1 \leq \ell \leq N} \max_{i,j \in G_\ell} \widehat{\Psi}_{ij,T} \leq q_{n,T}(\alpha)\right) \geq (1 - \alpha) + o(1).$$

Moreover, the same arguments as for Proposition 4.5 show that

$$\mathbb{P}\left(\min_{1 \leq \ell < \ell' \leq N} \min_{\substack{i \in G_\ell, \\ j \in G_{\ell'}}} \widehat{\Psi}_{ij,T} \leq q_{n,T}(\alpha)\right) = o(1).$$

Taken together, these two statements imply that

$$\mathbb{P}(B_{n,T}) \geq (1 - \alpha) + o(1). \quad (\text{C.34})$$

In what follows, we show that on the event $B_{n,T}$, (i) $\{\widehat{G}_1^{[n-N]}, \dots, \widehat{G}_N^{[n-N]}\} = \{G_1, \dots, G_N\}$ and (ii) $\widehat{N} = N$. From (i), (ii) and (C.34), the statements of Proposition 5.1 follow immediately.

Proof of (i). Suppose we are on the event $B_{n,T}$. The proof proceeds by induction on the iteration steps r of the HAC algorithm.

Base case ($r = 0$): In the first iteration step, the HAC algorithm merges two singleton clusters $\widehat{G}_i^{[0]} = \{i\}$ and $\widehat{G}_j^{[0]} = \{j\}$ with i and j belonging to the same group G_k . This is a direct consequence of (C.33). The algorithm thus produces a partition $\{\widehat{G}_1^{[1]}, \dots, \widehat{G}_{n-1}^{[1]}\}$ whose elements $\widehat{G}_\ell^{[1]}$ all have the following property: $\widehat{G}_\ell^{[1]} \subseteq G_k$ for some k , that is, each cluster $\widehat{G}_\ell^{[1]}$ contains elements from only one group.

Induction step ($r \rightsquigarrow r+1$): Now suppose we are in the r -th iteration step for some $r < n - N$. Assume that the partition $\{\widehat{G}_1^{[r]}, \dots, \widehat{G}_{n-r}^{[r]}\}$ is such that for any ℓ , $\widehat{G}_\ell^{[r]} \subseteq G_k$ for some k . Because of (C.33), the dissimilarity $\widehat{\Delta}(\widehat{G}_\ell^{[r]}, \widehat{G}_{\ell'}^{[r]})$ gets minimal for two clusters $\widehat{G}_\ell^{[r]}$ and $\widehat{G}_{\ell'}^{[r]}$

with the property that $\widehat{G}_\ell^{[r]} \cup \widehat{G}_{\ell'}^{[r]} \subseteq G_k$ for some k . Hence, the HAC algorithm produces a partition $\{\widehat{G}_1^{[r+1]}, \dots, \widehat{G}_{n-(r+1)}^{[r+1]}\}$ whose elements $\widehat{G}_\ell^{[r+1]}$ are all such that $\widehat{G}_\ell^{[r+1]} \subseteq G_k$ for some k .

The above induction argument shows the following: For any $r \leq n - N$, the partition $\{\widehat{G}_1^{[r]}, \dots, \widehat{G}_{n-r}^{[r]}\}$ consists of clusters $\widehat{G}_\ell^{[r]}$ which all have the property that $\widehat{G}_\ell^{[r]} \subseteq G_k$ for some k . This in particular holds for the partition $\{\widehat{G}_1^{[n-N]}, \dots, \widehat{G}_N^{[n-N]}\}$, which implies that $\{\widehat{G}_1^{[n-N]}, \dots, \widehat{G}_N^{[n-N]}\} = \{G_1, \dots, G_N\}$. $\square$

Proof of (ii). To start with, consider any partition $\{\widehat{G}_1^{[n-r]}, \dots, \widehat{G}_r^{[n-r]}\}$ with $r < N$ elements. Such a partition must contain at least one element $\widehat{G}_\ell^{[n-r]}$ with the following property: $\widehat{G}_\ell^{[n-r]} \cap G_k \neq \emptyset$ and $\widehat{G}_\ell^{[n-r]} \cap G_{k'} \neq \emptyset$ for some $k \neq k'$. On the event $B_{n,T}$, it obviously holds that $\widehat{\Delta}(S) > q_{n,T}(\alpha)$ for any S with the property that $S \cap G_k \neq \emptyset$ and $S \cap G_{k'} \neq \emptyset$ for some $k \neq k'$. Hence, we can infer that on the event $B_{n,T}$, $\max_{1 \leq \ell \leq r} \widehat{\Delta}(\widehat{G}_\ell^{[n-r]}) > q_{n,T}(\alpha)$ for any $r < N$.

Next consider the partition $\{\widehat{G}_1^{[n-r]}, \dots, \widehat{G}_r^{[n-r]}\}$ with $r = N$ and suppose we are on the event $B_{n,T}$. From (i), we already know that $\{\widehat{G}_1^{[n-N]}, \dots, \widehat{G}_N^{[n-N]}\} = \{G_1, \dots, G_N\}$. Moreover, $\widehat{\Delta}(G_\ell) \leq q_{n,T}(\alpha)$ for any ℓ . Hence, we obtain that $\max_{1 \leq \ell \leq N} \widehat{\Delta}(\widehat{G}_\ell^{[n-N]}) = \max_{1 \leq \ell \leq N} \widehat{\Delta}(G_\ell) \leq q_{n,T}(\alpha)$.

Putting everything together, we can conclude that on the event $B_{n,T}$,

$$\min \left\{ r = 1, 2, \dots \mid \max_{1 \leq \ell \leq r} \widehat{\Delta}(\widehat{G}_\ell^{[n-r]}) \leq q_{n,T}(\alpha) \right\} = N,$$

that is, $\widehat{N} = N$. $\square$

Proof of Proposition 5.2

We consider the event

$$D_{n,T} = \left\{ \widehat{\Phi}_{n,T} \leq q_{n,T}(\alpha) \text{ and } \min_{1 \leq \ell < \ell' \leq N} \min_{\substack{i \in G_\ell, \\ j \in G_{\ell'}}} \widehat{\Psi}_{ij,T} > q_{n,T}(\alpha) \right\},$$

where we write the statistic $\widehat{\Phi}_{n,T}$ as

$$\widehat{\Phi}_{n,T} = \max_{1 \leq i < j \leq n} \max_{(u,h) \in \mathcal{G}_T} \left\{ \left| \frac{\widehat{\psi}_{ij,T}(u,h) - \widehat{\psi}_{ij,T}^{\text{trend}}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| - \lambda(h) \right\}$$

with $\widehat{\psi}_{ij,T}^{\text{trend}}(u,h) = \sum_{t=1}^T w_{t,T}(u,h) \{(m_{i,T}(t/T) - \bar{m}_{i,T}) - (m_{j,T}(t/T) - \bar{m}_{j,T})\}$ and $\bar{m}_{i,T} = T^{-1} \sum_{t=1}^T m_{i,T}(t/T)$. The event $D_{n,T}$ can be analysed by the same arguments as those applied to the event $B_{n,T}$ in the proof of Proposition 5.1. In particular, analogous to (C.34) and statements (i) and (ii) in this proof, we can show that

$$\mathbb{P}(D_{n,T}) \geq (1 - \alpha) + o(1) \quad (\text{C.35})$$

and

$$D_{n,T} \subseteq \{\widehat{N} = N \text{ and } \widehat{G}_\ell = G_\ell \text{ for all } \ell\}. \quad (\text{C.36})$$

Moreover, we have that

$$D_{n,T} \subseteq \bigcap_{1 \leq \ell < \ell' \leq \widehat{N}} E_{n,T}^{[\ell, \ell']}(\alpha), \quad (\text{C.37})$$

which is a consequence of the following observation: For all i, j and $(u, h) \in \mathcal{G}_T$ with

$$\left| \frac{\widehat{\psi}_{ij,T}(u,h) - \widehat{\psi}_{ij,T}^{\text{trend}}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| - \lambda(h) \leq q_{n,T}(\alpha) \quad \text{and} \quad \left| \frac{\widehat{\psi}_{ij,T}(u,h)}{(\widehat{\sigma}_i^2 + \widehat{\sigma}_j^2)^{1/2}} \right| - \lambda(h) > q_{n,T}(\alpha),$$

it holds that $\widehat{\psi}_{ij,T}^{\text{trend}}(u,h) \neq 0$, which in turn implies that $m_i(v) - m_j(v) \neq 0$ for some $v \in I_{u,h}$. From (C.36) and (C.37), we obtain that

$$D_{n,T} \subseteq \left\{ \bigcap_{1 \leq \ell < \ell' \leq \widehat{N}} E_{n,T}^{[\ell, \ell']}(\alpha) \right\} \cap \{\widehat{N} = N \text{ and } \widehat{G}_\ell = G_\ell \text{ for all } \ell\} = E_{n,T}(\alpha).$$

This together with (C.35) implies that $\mathbb{P}(E_{n,T}(\alpha)) \geq (1 - \alpha) + o(1)$.

References

- BARRO, R. J. and LEE, J. W. (2013). A new data set of educational attainment in the world, 1950–2010. *Journal of Development Economics*, **104** 184–198.
- BERKES, I., LIU, W. and WU, W. B. (2014). Komlós-Major-Tusnády approximation under dependence. *Annals of Probability*, **42** 794–817.
- CARLSTEIN, E. (1986). The use of subseries values for estimating the variance of a general statistic from a stationary sequence. *Annals of Statistics*, **14** 1171–1179.

- CHERNOZHUKOV, V., CHETVERIKOV, D. and KATO, K. (2017). Central limit theorems and bootstrap in high dimensions. *Annals of Probability*, **45** 2309–2352.
- KHISMATULLINA, M. and VOGT, M. (2020). Multiscale inference and long-run variance estimation in non-parametric regression with time series errors. *Journal of the Royal Statistical Society: Series B*, **82** 5–37.
- LEE, C.-C. (2005). Energy consumption and GDP in developing countries: a cointegrated panel analysis. *Energy Economics*, **27** 415–427.
- LEE, C.-H. and HUANG, B.-N. (2002). The relationship between exports and economic growth in East Asian countries: a multivariate threshold autoregressive approach. *Journal of Economic Development*, **27** 45–68.
- NAZAROV, F. (2003). On the maximal perimeter of a convex set in $\mathbb{R}^n$ with respect to a Gaussian measure. In *Geometric Aspects of Functional Analysis*. Springer, 169–187.
- PARK, C., VAUGHAN, A., HANNIG, J. and KANG, K.-H. (2009). SiZer analysis for the comparison of time series. *Journal of Statistical Planning and Inference*, **139** 3974–3988.
- SAX, C. and EDDELBUETTEL, D. (2018). Seasonal adjustment by X-13ARIMA-SEATS in R. *Journal of Statistical Software*, **87** 1–17.
- SHARMA, S. C. and DHAKAL, D. (1994). Causal analyses between exports and economic growth in developing countries. *Applied Economics*, **26** 1145–1157.
- WU, W. B. and WU, Y. N. (2016). Performance bounds for parameter estimates of high-dimensional linear models with correlated errors. *Electronic Journal of Statistics*, **10** 352–379.
- WU, W. B. and ZHAO, Z. (2007). Inference of trends in time series. *Journal of the Royal Statistical Society: Series B*, **69** 391–410.
- ZHANG, Y., SU, L. and PHILLIPS, P. C. (2012). Testing for common trends in semi-parametric panel data models with fixed effects. *The Econometrics Journal*, **15** 56–100.